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Prime number theorem and Dirichlet's theorem

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Alla mia famiglia

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Abstract

This dissertation presents the proofs of two classical theorems of analytic number theory: the prime number theorem and Dirichlet's theorem on arithmetic progressions. The prime number theorem states that the number of primes less than x is asymptotic to $x/\log x$ as $x \rightarrow +\infty$. Dirichlet's theorem on arithmetic progressions says that given m, a coprime integers there exist infinitely many primes $p \equiv a \pmod{m}$. Moreover, let $\mathbb{P}_a$ be the set of prime numbers such that $p \equiv a \pmod{m}$. The set $\mathbb{P}_a$ has logarithmic density $1/\varphi(m)$.

In the introductory chapter we describe some fundamental notions and tools necessary to understand the rest of the dissertation. In particular, we introduce several classical arithmetic functions, state some of their basic properties, and give the definition of Dirichlet formal series. In the final part of the chapter we collect and prove various basic results, such as Abel's lemma on summation by parts and Mertens' theorems, which will be used several times in the following chapters.

In chapter 1 we explain Zagier's proof of the prime number theorem [6], expanding on the details of his approach. In particular, we analyze the properties of some arithmetic functions that play a fundamental role in analytic number theory, such as Riemann's zeta function $\zeta(s) = \sum_{n \in \mathbb{N}} 1/n^s$, $\vartheta(x) = \sum_{p \leq x} \log p$, and $\Phi(s) = \sum_{p \in \mathbb{P}} \log p/p^s$. The proof proposed by Zagier in his article relies on two main ingredients: showing that Riemann's zeta function $\zeta(s)$ does not vanish at s for $\Re(s) = 1$ and deducing the prime number theorem from this.

In chapter 2 we understand how natural, logarithmic, and analytic densities are related to each other for a generic subset consisting of prime numbers. First of all, we prove the equivalence between logarithmic and analytic density. Subsequently, we show that a subset that admits natural density also admits logarithmic density, and the values agree. Showing an appropriate example, we prove that the opposite implication is not valid. In particular, the set considered consists of prime numbers that begin with 1 in decimal representation.

In the last chapter, we provide proof on Dirichlet's theorem of arithmetic progressions by retracing the path followed by Serre in his book [4]. First of all, we introduce Dirichlet characters. They acquire a relevant role in algebraic

number theory. We define the series $L(s, \chi)$ thanks to Dirichlet characters and analyze their properties. Finally we prove Dirichlet's theorem, which not only guarantees the infinitude of the primes $p \equiv a \pmod{m}$ but allows us to establish that the set consisting of these primes has a logarithmic density equal to $1/\varphi(m)$.

Preliminaries

In this chapter we review some fundamental facts in number theory that will prove useful for a better understanding of the dissertation.

More specifically, we will define and study the main properties of arithmetic functions and formal Dirichlet series. Finally, we will list and study some of the most important arithmetic functions, describe the first Möbius inversion formula, Abel's lemma on summation by parts and Mertens' theorems.

0.1 Arithmetic functions

In the dissertation we will use the notation $\mathbb{N}$ to indicate the set of positive integers and $\mathbb{P}$ to refer to the set of prime numbers.

Definition 0.1. An arithmetic function is any function

$$f : \mathbb{N} \rightarrow X \subseteq \mathbb{C}.$$

We will say that an arithmetic function is multiplicative if it is not identically zero and, for every $m, n \in \mathbb{N}$,

$$(m, n) = 1 \Rightarrow f(mn) = f(m)f(n),$$

while we will say that it is a completely multiplicative function if it is not identically zero and, for every $m, n \in \mathbb{N}$,

$$f(mn) = f(m)f(n).$$

Similarly, we will say that an arithmetic function is additive if it is not identically zero and, for every $m, n \in \mathbb{N}$,

$$(m, n) = 1 \Rightarrow f(mn) = f(m) + f(n),$$

and that it is a completely additive function if it is not identically zero and, for every $m, n \in \mathbb{N}$,

$$f(mn) = f(m) + f(n).$$

For example, the functions

$$\begin{aligned} \mathbf{e} : \mathbb{N} &\rightarrow \{0, 1\} \\ n &\mapsto \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases} \end{aligned}$$

$$\begin{aligned} \mathbf{1} : \mathbb{N} &\rightarrow \{1\} \\ n &\mapsto 1, \end{aligned}$$

and, for every $\alpha \in \mathbb{C}$,

$$\begin{aligned} i^\alpha : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto n^\alpha \end{aligned}$$

are completely multiplicative functions (in particular the identity on $\mathbb{N}$ is a completely multiplicative function). Instead, the logarithm function

$$\begin{aligned} L : \mathbb{N} &\rightarrow \mathbb{R} \\ n &\mapsto \log n \end{aligned}$$

is completely additive. We will denote by $\mathbb{A}$ the set of the arithmetic functions.

0.2 Formal Dirichlet series

We will call **formal Dirichlet series** a series of the form

$$\sum_{n \in \mathbb{N}} f(n) \omega(n),$$

where f is an arithmetic function and ω is a completely multiplicative indeterminate. We will denote by $\mathbb{S}$ the set of formal Dirichlet series.

To make the notation easier to follow we will use capital letters to indicate the formal Dirichlet series, while the corresponding lowercase letters will indicate the associated arithmetic function.

In $\mathbb{S}$ we can define two operations of sum and product: if F and G are two formal Dirichlet series, we define their sum and their product as

$$F + G = \sum_{n \in \mathbb{N}} (f(n) + g(n)) \omega(n),$$

$$FG = \sum_{n \in \mathbb{N}} \left(\sum_{hk=n} f(k)g(h) \right) \omega(n).$$

We can observe that $(\mathbb{S}, +, \cdot)$ is a commutative and unitary ring. The product of two formal series can be written again in the following way:

$$FG = \sum_{n \in \mathbb{N}} \left(\sum_{d|n} f(d)g\left(\frac{n}{d}\right) \right) \omega(n).$$

Let f and g be two arithmetic functions. The **Dirichlet convolution** between f and g is the arithmetic function

$$\begin{aligned} f * g : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto \sum_{d|n} f(d)g\left(\frac{n}{d}\right), \end{aligned}$$

so we can rewrite

$$FG = \sum_{n \in \mathbb{N}} (f * g)(n) \omega(n).$$

We now list without proof a series of properties of the algebraic structure of the rings $\mathbb{A}$ and $\mathbb{S}$.

We can define the **norm** application on $\mathbb{S}$:

$$\nu : \mathbb{S} \rightarrow \mathbb{N} \cup \{+\infty\}$$

$$\sum_{n \in \mathbb{N}} f(n)\omega(n) \mapsto \begin{cases} \min\{n \in \mathbb{N} \mid f(n) \neq 0\} & \text{if } F \neq 0, \\ +\infty & \text{if } F = 0. \end{cases}$$

Proposition 0.2. *If F and G are two formal Dirichlet series, then $\nu(FG) = \nu(F)\nu(G)$. In particular, $\mathbb{S}$ is an integral domain.*

Theorem 0.3. *The Dirichlet convolution is associative and has a neutral element (the arithmetic function $\mathbf{e}$).*

Corollary 0.4. *$(\mathbb{A}, +, *)$ is a commutative and unitary ring. Moreover, $\mathbb{A}$ and $\mathbb{S}$ are isomorphic (as rings) and an isomorphism is given by the map*

$$\mathbb{A} \rightarrow \mathbb{S}$$

$$f \mapsto \sum_{n \in \mathbb{N}} f(n)\omega(n).$$

We will denote by $\mathbf{1}$ and ζ the formal Dirichlet series associated with $\mathbf{e}$ and $\mathbf{1}$ respectively:

$$\mathbf{1} = \sum_{n \in \mathbb{N}} \mathbf{e}(n)\omega(n) = \omega(1),$$

$$\zeta = \sum_{n \in \mathbb{N}} \mathbf{1}(n)\omega(n) = \sum_{n \in \mathbb{N}} \omega(n).$$

An arithmetic function f is **invertible** if there exists an arithmetic function g such that

$$f * g = \mathbf{e}.$$

Furthermore, we will say that a formal series F is **invertible** if the associated arithmetic function is invertible. Invertible formal series form a multiplicative subgroup of $\mathbb{S}$, that will be denoted by $\mathbb{U}$.

0.3 Multiplicative functions

In this section we will list some general properties of multiplicative functions. Some of these facts will be proven. Let us remark that, clearly, every completely multiplicative function is a multiplicative function.

Proposition 0.5. *Let f be an arithmetic function. If f is multiplicative, then $f(1) = 1$.*

Proof. Let a be a positive integer with $f(a) \neq 0$. Then

$$f(a) = f(a \cdot 1) = f(a)f(1) \Rightarrow f(1) = 1.$$

□

We will denote the set of multiplicative functions by $\mathbb{M}$.

Lemma 0.6. *$(\mathbb{M}, *)$ is a group.*

A proof of this fact can be found in [2, Proposition 6.7 in Chapter VI].

Corollary 0.7. *Let f and g two arithmetic functions such that f and $f * g$ are multiplicative functions. Then g is also a multiplicative function.*

If an arithmetic function is multiplicative, to determine it completely on all of $\mathbb{N}$ it is enough to know its behavior on the powers of the primes (and if it is completely multiplicative it is enough to know its behavior on the primes).

Theorem 0.8. *Let $\mathcal{P} = \{p^m \mid p \in \mathbb{P}, m \in \mathbb{N}\}$ and f be a multiplicative function. If*

$$\lim_{\substack{x \rightarrow +\infty \\ x \in \mathcal{P}}} f(x) = 0,$$

then

$$\lim_{\substack{n \rightarrow +\infty \\ n \in \mathbb{N}}} f(n) = 0.$$

Proof. By hypothesis, there exists a positive constant B such that

$$\forall p^m \in \mathcal{P}, p^m > B \Rightarrow |f(p^m)| \leq 1.$$

Since the set $\{(p, m) \mid p^m \leq B\}$ is finite, there exists another positive constant A such that

$$\forall p \in \mathcal{P}, \forall m \geq 1 \quad |f(p^m)| \leq A.$$

Furthermore, for every $\varepsilon > 0$ there exists a positive constant $N(\varepsilon)$ such that

$$\forall p \in \mathbb{P}, \forall m \in \mathbb{N}, \quad p^m > N(\varepsilon) \Rightarrow |f(p^m)| < \varepsilon.$$

The constants A, B are independent of the choice of p, m and ε . Moreover, the sets

$$\mathcal{V}_\varepsilon = \{n \in \mathbb{N} \mid \forall p \in \mathbb{P}, \forall m \in \mathbb{N}, p^m \parallel n \Rightarrow p^m \leq N(\varepsilon)\},$$

$$\mathcal{W} = \{(p, m) \in \mathbb{P} \times \mathbb{N} \mid p^m \leq B\}$$

are necessarily finite: let $\rho(\varepsilon)$ be the maximum of $\mathcal{V}_\varepsilon$, $C = |\mathcal{W}|$; we can observe that C is also independent of p, m and of the choice of ε . Let n be a positive integer with $n > \rho(\varepsilon)$: then there exist a prime $\hat{p}$ and a positive integer m such that $\hat{p}^m \parallel n$ and $\hat{p}^m > N(\varepsilon)$. Writing

$$n = \hat{p}^m \prod_{(p,a) \notin \mathcal{W}} p^a \prod_{(q,b) \in \mathcal{W}} q^b$$

we obtain

$$|f(n)| = |f(\hat{p}^m)| \prod_{(p,a) \notin \mathcal{W}} |f(p^a)| \prod_{(q,b) \in \mathcal{W}} |f(q^b)| < A^C \varepsilon,$$

and since this is true for every choice of $n > \rho(\varepsilon)$, the claim follows. $\square$

0.4 Additive functions

Obviously, every completely additive function is an additive function.

Proposition 0.9. *Let f be an arithmetic function. If f is an additive function, then $f(1) = 0$.*

Proof. Let a be a positive integer. We have

$$f(a) = f(a \cdot 1) = f(a) + f(1) \Rightarrow f(1) = 0.$$

$\square$

As well as for multiplicative functions, if an arithmetic function is additive, to determine it completely on all of $\mathbb{N}$ it is enough to know its behavior on the powers of the primes (and if it is completely additive it is enough to know its behavior on the primes).

0.5 Main arithmetic functions

Divisor functions, d_r

We will call Dirichlet divisor function the arithmetic function

$$d = \mathbf{1} * \mathbf{1} : \mathbb{N} \rightarrow \mathbb{N}$$

$$n \mapsto \sum_{d|n} 1.$$

Proposition 0.10. *The arithmetic function d is multiplicative, but not completely multiplicative.*

Proof. Since $\mathbb{M}$ is a group and $\mathbf{1} \in \mathbb{M}$, the function d is multiplicative, but not completely multiplicative: indeed, if p is a prime number, $d(p) = 2$, $d(p^2) = 3$ and $d(p^2) \neq d(p)^2$. □

More generally, we can take the convolution of the arithmetic function $\mathbf{1}$ with itself finitely many times: the arithmetic function

$$d_r = *_{i=1}^r \mathbf{1} : \mathbb{N} \rightarrow \mathbb{N}$$

$$n \mapsto \sum_{d_1 \dots d_r = n} 1$$

counts the number of ways a positive integer n can be written as a product of r positive integers (for example $d = d_2$). Since the Dirichlet divisor function is multiplicative, to determine it completely it is enough to study its behavior on the powers of primes: it is easy to observe that

$$d(p^a) = a + 1,$$

and consequently

Proposition 0.11. *For every positive integer n ,*

$$d(n) = \prod_{p^a || n} (a + 1).$$

Corollary 0.12. *$d(n)$ is odd if and only if n is a square.*

Möbius function, μ

First of all we define the functions ω, Ω :

$$\begin{aligned} \omega : \mathbb{N} &\rightarrow \mathbb{N} \\ n &\mapsto \begin{cases} 0 & \text{if } n = 1, \\ \sum_{p|n} 1 & \text{if } n \neq 1 \end{cases} \end{aligned}$$

$$\begin{aligned} \Omega : \mathbb{N} &\rightarrow \mathbb{N} \\ n &\mapsto \begin{cases} 0 & \text{if } n = 1, \\ \sum_{p^a || n} a & \text{if } n \neq 1 \end{cases} \end{aligned}$$

that count, respectively, the number of primes that appear in the factorization of a positive integer and the sum of the exponents of these primes.

Moreover, we can define the **radical** function that associates each positive integer with its largest square free divisor:

$$\begin{aligned} \rho : \mathbb{N} &\rightarrow \mathbb{N} \\ n &\mapsto \begin{cases} 1 & \text{if } n = 1, \\ \prod_{p|n} p & \text{if } n \neq 1. \end{cases} \end{aligned}$$

The functions ω and Ω are not multiplicative, but it is easy to observe that ω is additive and Ω is completely additive; furthermore $\Omega(n) = \omega(n)$ if and only if n is square-free.

The **Möbius function** is the arithmetic function

$$\begin{aligned} \mu : \mathbb{N} &\rightarrow \{-1, 0, 1\} \\ n &\mapsto \begin{cases} 1 & \text{if } n = 1, \\ (-1)^r & \text{if } \Omega(n) = \omega(n) = r, \\ 0 & \text{if } \Omega(n) > \omega(n). \end{cases} \end{aligned}$$

Proposition 0.13. μ is a multiplicative function and, more specifically, μ is an invertible arithmetic function.

Proof. Let m, n be two coprime positive integers. If at least one of them is not square-free, neither is mn . So $\mu(mn) = 0$ and at least one of $\mu(n)$ or $\mu(m)$ is 0, therefore the equality $\mu(mn) = \mu(m)\mu(n)$ holds. We may thus assume that m and n are square-free. Then so is mn since $(m, n) = 1$. Thus if m has k different prime factors and n has l different prime factors, mn has $l + k$ different prime factors, since the function ω is additive. Hence the following holds $\mu(m)\mu(n) = (-1)^k(-1)^l = (-1)^{k+l} = \mu(mn)$. $\square$

The μ function owes its importance to the following result, known as **first Möbius inversion formula**:

Theorem 0.14 (Möbius).

$$\mu * \mathbf{1} = \mathbf{e},$$

that is

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n \neq 1. \end{cases}$$

Moreover, if f and g are arithmetic functions, then

$$f * \mathbf{1} = g \Leftrightarrow f = g * \mu.$$

Proof. Let n be a positive integer. We can observe that $\mu * \mathbf{1}(1) = \mu(1) = 1$, whereas if $n > 1$, we can note that the divisors d of n such that $\mu(d) \neq 0$ are the square-free ones. Consequently, if we set $\rho(n) = q_1 \dots q_r$, we get

$$\sum_{d|n} \mu(d) = \sum_{d|\rho(n)} \mu(d).$$

Now the divisors of $\rho(n)$ are all and only the possible products of the primes that appear in its factorization. Let k be a integer with $0 \leq k \leq r$. There exist $\binom{r}{k}$ divisors that are product of k of these primes. Hence,

$$\sum_{d|\rho(n)} \mu(d) = \sum_{k=0}^r \binom{r}{k} (-1)^k = (1 + (-1))^r = 0.$$

Now we prove the second part of the theorem:

$$(\Rightarrow) g = f * \mathbf{1} \Rightarrow g * \mu = f * \mathbf{1} * \mu = f * (\mathbf{1} * \mu) = f * \mathbf{e} = f.$$

$$(\Leftarrow) f = g * \mu \Rightarrow f * \mathbf{1} = g * \mu * \mathbf{1} = g * (\mu * \mathbf{1}) = g * \mathbf{e} = g.$$

$\square$

We can associate with the arithmetic function μ the formal Dirichlet series

$$\zeta^{-1} = \frac{1}{\zeta} = \sum_{n \in \mathbb{N}} \mu(n) \omega(n).$$

The first Möbius inversion formula has the following corollary:

Corollary 0.15. *Let f be an arithmetic function. If f is completely multiplicative, then its inverse function is $f^{-1} = f\mu$.*

Euler's totient function, φ

Euler's function is :

$$\begin{aligned} \varphi : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto \sum_{\substack{1 \leq k \leq n \\ (k,n)=1}} 1. \end{aligned}$$

Proposition 0.16. $\varphi = \mu * i$. In particular, Euler's function φ is multiplicative.

This result is proven in [3, Theorem 262 Chapter XVI]. By using the first Möbius inversion formula, we obtain the following corollary:

Corollary 0.17. $i = \varphi * \mathbf{1}$, that is, for every positive integer n we have that

$$\sum_{d|n} \varphi(d) = n.$$

Since φ is completely multiplicative, to determine its behavior it is sufficient to evaluate it on the powers of primes. Consequently, the following proposition holds:

Proposition 0.18. For every positive integer n ,

$$\varphi(n) = \prod_{p^a || n} p^{a-1}(p-1) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

This equation is proven in [3, Theorem 62 in Chapter V].

von Mangoldt function, Λ

We will call **von Mangoldt function** the arithmetic function

$$\begin{aligned} \Lambda : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto \begin{cases} 0 & \text{if } n = 1 \vee \omega(n) \geq 2, \\ \log p & \text{if } n = p^a, a \geq 1. \end{cases} \end{aligned}$$

Proposition 0.19. $\Lambda * \mathbf{1} = L$ and $\Lambda = L * \mu$.

Proof. Let $n > 1$ be a positive integer. Since Λ is nonzero only if it is evaluated at a power of prime,

$$\sum_{d|n} \Lambda(d) = \sum_{p^a|n} \Lambda(p^a) = \sum_{p|n} \sum_{a:p^a|n} \Lambda(p^a) = \sum_{p^a||n} \log p^a = \log \left(\prod_{p^a||n} p^a \right) = \log n.$$

By using the first Möbius inversion formula, we have $\Lambda = L * \mu$. □

Sum of divisors function, σ

We will call **sum of divisors function** the arithmetic function

$$\begin{aligned} \sigma &:= i * \mathbf{1} : \mathbb{N} \rightarrow \mathbb{N} \\ n &\mapsto \sum_{d|n} d. \end{aligned}$$

σ is multiplicative too, since it is the convolution of multiplicative functions. If we evaluate it on the powers of prime, then we obtain

$$\sigma(p^a) = \sum_{i=0}^a p^i = \frac{p^{a+1} - 1}{p - 1},$$

and consequently

Proposition 0.20. *For every positive integer n ,*

$$\sigma(n) = \prod_{p^a||n} \frac{p^{a+1} - 1}{p - 1}.$$

0.6 Dirichlet series

Definition 0.21. Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence of complex numbers and s a complex number. We define **Dirichlet series** associated with the sequence $\{a_n\}_{n \in \mathbb{N}}$ and s the series

$$\sum_{n \in \mathbb{N}} \frac{a_n}{n^s}.$$

To connect with the previously given definition of Dirichlet's formal series we will take

$$\begin{aligned} f : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto a_n \end{aligned}$$

and, as completely multiplicative indeterminate,

$$\begin{aligned} \omega : \mathbb{N} &\rightarrow \mathbb{C} \\ n &\mapsto n^{-s}. \end{aligned}$$

Definition 0.22. The **abscissa of convergence** of a Dirichlet series σ_c can be defined as

$$\sigma_c = \inf \left\{ \sigma \in \mathbb{R} : \sum_{n=1}^{+\infty} \frac{a_n}{n^s} \text{ converges for every } s \text{ for which } \Re(s) > \sigma \right\}.$$

The line $\Re(s) = \sigma_c$ is called the **line of convergence**. The **half-plane of convergence** is defined as

$$\mathbb{C}_{\sigma_c} = \{s \in \mathbb{C} : \Re(s) > \sigma_c\}.$$

By abuse of language we consider $\emptyset$ and $\mathbb{C}$ as open half planes. The cases $\emptyset$ and $\mathbb{C}$ correspond respectively to $\sigma_c = +\infty$ and $\sigma_c = -\infty$.

Lemma 0.23 (Abel). *Let $\{x_n\}_{n \in \mathbb{N}}$ and $\{a_n\}_{n \in \mathbb{N}}$ be two sequences of real numbers with the first of them increasing, taking non-negative values and such that*

$$\lim_{n \rightarrow +\infty} x_n = +\infty.$$

Moreover, let $f : (0, +\infty) \rightarrow \mathbb{C}$ be a function and

$$A : [0, +\infty) \rightarrow \mathbb{R}$$

$$x \mapsto \sum_{\substack{n \in \mathbb{N} \\ x_n \leq x}} a_n.$$

The following holds

$$\sum_{k=1}^n a_k f(x_k) = A(x_n) f(x_n) - \sum_{k=1}^{n-1} A(x_k) (f(x_{k+1}) - f(x_k)).$$

Furthermore, if $f \in \mathcal{C}^1(0, +\infty)$,

$$\sum_{\substack{n \in \mathbb{N} \\ x_n \leq x}} a_n f(x_n) = A(x) f(x) - \int_{x_1}^x A(u) f'(u) du,$$

and, more specifically,

$$\sum_{n \in \mathbb{N}} a_n f(n) = A(x) f(x) - \int_1^x A(u) f'(u) du.$$

Proof. We set, by convention, $A(x_0) = 0$. By definition we have $a_k = A(x_k) - A(x_{k-1})$, hence we get

$$\begin{aligned} \sum_{k=1}^n a_k f(x_k) &= \sum_{k=1}^n (A(x_k) - A(x_{k-1})) f(x_k) \\ &= A(x_n) f(x_n) - \sum_{k=1}^{n-1} A(x_k) (f(x_{k+1}) - f(x_k)), \end{aligned}$$

and the first claim follows.

Now we prove the second statement: we can observe that A is constant in every interval (x_k, x_{k+1}) . Let n be the greatest positive integer such as $x_n \leq x$: we have

$$\begin{aligned}
 \sum_{k=1}^n a_k f(x_k) &= A(x_n) f(x_n) - \sum_{k=1}^{n-1} A(x_k) (f(x_{k+1}) - f(x_k)) \\
 &= A(x_n) f(x_n) - \sum_{k=1}^{n-1} A(x_k) \int_{x_k}^{x_{k+1}} f'(u) du \\
 &= A(x) f(x) - \int_{x_n}^x A(u) f'(u) du - \sum_{k=1}^{n-1} A(x_k) \int_{x_k}^{x_{k+1}} f'(u) du \\
 &= A(x) f(x) - \int_{x_1}^x A(u) f'(u) du.
 \end{aligned}$$

The third claim follows immediately from the second one by taking $x_k = k$ for every k . □

Definition 0.24. Given an arithmetic function f we will call

$$\begin{aligned}
 F : \mathbb{C}_\sigma &\rightarrow \mathbb{C} \\
 s &\mapsto \sum_{n \in \mathbb{N}} \frac{f(n)}{n^s},
 \end{aligned}$$

where σ is the abscissa of convergence, the **generating function** of the arithmetic function f .

Proposition 0.25. *Let F be the generating function of f , for s such that $\operatorname{Re}(s) > \sigma_1$ and G be the generating function of g , for s such that $\operatorname{Re}(s) > \sigma_2$. $F + G$ coincides with the generating function of $f + g$ on the set $\{s \in \mathbb{C} \mid \operatorname{Re}(s) > \sigma = \max\{\sigma_1, \sigma_2\}\}$, while FG coincides with the generating function of $f * g$, for s such that $\operatorname{Re}(s) > \max\{\sigma_1, \sigma_2\}$.*

A proof of this proposition can be found in [2, Theorem 6.9 in Chapter VI].

For instance, the generating function of the arithmetic function **1** is the **Riemann zeta function**

$$\begin{aligned}
 \zeta : \mathbb{C}_\sigma &\rightarrow \mathbb{C} \\
 s &\mapsto \sum_{n \in \mathbb{N}} \frac{1}{n^s},
 \end{aligned}$$

whose abscissa of convergence is $\sigma = 1$.

Definition 0.26. Let $\{a_n\}$ be a sequence of complex numbers. We call **infinite product** associated with the sequence $\{a_n\}$ the quantity

$$\prod_{n=1}^{+\infty} a_n = \lim_{N \rightarrow +\infty} \prod_{n=1}^N a_n,$$

if the latter limit exists.

Theorem 0.27 (Euler). Let f be a multiplicative function and σ be the abscissa of convergence of the Dirichlet series associated with it. For every s such that $\operatorname{Re}(s) > \sigma$,

$$\sum_{n \in \mathbb{N}} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{hs}},$$

and if f is completely multiplicative

$$\sum_{n \in \mathbb{N}} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(1 - \frac{f(p)}{p^s}\right)^{-1}.$$

In particular,

$$\zeta(s) = \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p^s}\right)^{-1}.$$

Proof. Let N be a positive integer and consider the truncation

$$\prod_{p \leq N} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{sh}}.$$

By using the multiplicativity of f , we can observe that the previous equation can be written in the following way

$$\prod_{p \leq N} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{sh}} = \sum_{\substack{n \in \mathbb{N} \\ p|n \Rightarrow p \leq N}} \frac{f(n)}{n^s}. \quad (1)$$

Every natural number less than or equal to N can be factored by using primes less than or equal to N : therefore we can break the summation in the following way

$$\sum_{\substack{n \in \mathbb{N} \\ p|n \Rightarrow p \leq N}} \frac{f(n)}{n^s} = \sum_{n=1}^N \frac{f(n)}{n^s} + \sum_{\substack{n > N \\ p|n \Rightarrow p \leq N}} \frac{f(n)}{n^s}. \quad (2)$$

Combining (1) and (2) we obtain

$$\left| \prod_{p \leq N} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{sh}} - \sum_{n=1}^N \frac{f(n)}{n^s} \right| \leq \sum_{\substack{n > N \\ p|n \Rightarrow p \leq N}} \frac{|f(n)|}{n^s} \leq \sum_{n > N} \frac{|f(n)|}{n^s},$$

and since the latter is a tail of a convergent series we obtain

$$\lim_{N \rightarrow +\infty} \left| \prod_{p \leq N} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{sh}} - \sum_{n=1}^N \frac{f(n)}{n^s} \right| = 0 \Rightarrow \prod_{p \in \mathbb{P}} \sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{hs}} = \sum_{n \in \mathbb{N}} \frac{f(n)}{n^s}.$$

If the function f is completely multiplicative,

$$\sum_{h=0}^{+\infty} \frac{f(p^h)}{p^{sh}} = \sum_{h=0}^{+\infty} \frac{f(p)^h}{p^{sh}} = \sum_{h=0}^{+\infty} \left(\frac{f(p)}{p^s} \right)^h.$$

We observe that, from the definition of abscissa of convergence, for $\Re(s) > \sigma$, the terms of the series $f(p^h)/p^{hs}$ tend to 0 in absolute value whereby $|f(p)/p^s|^h \rightarrow 0$. But this happens if and only if

$$\left| \frac{f(p)}{p^s} \right| < 1,$$

therefore

$$\sum_{h=0}^{+\infty} \left(\frac{f(p)}{p^s} \right)^h = \frac{1}{1 - \frac{f(p)}{p^s}} = \left(1 - \frac{f(p)}{p^s} \right)^{-1},$$

whence the second statement. The last claim follows immediately from the second one considering as arithmetic function the completely multiplicative function **1**.

□

0.7 Euler's constant and Mertens' theorems

Theorem 0.28. *There exists a real number γ , called **Euler's constant**, such that $0 < \gamma < 1$ and the following asymptotics holds:*

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + \mathcal{O}\left(\frac{1}{x}\right) \quad (\text{as } x \rightarrow +\infty).$$

Proof. We apply Abel's lemma on summation by parts, with the sequence $\{\frac{1}{n}\}_{n \in \mathbb{N}}$ and the function

$$f : (0, +\infty) \rightarrow \mathbb{R} \\ x \mapsto \frac{1}{x}.$$

We obtain

$$\begin{aligned} \sum_{n \leq x} \frac{1}{n} &= \frac{1}{x} \sum_{n \leq x} 1 - \int_1^x \left(-\frac{1}{u^2}\right) \sum_{n \leq u} 1 \, du \\ &= \frac{\lfloor x \rfloor}{x} + \int_1^x \frac{\lfloor u \rfloor}{u^2} \, du = 1 - \frac{\{x\}}{x} + \int_1^x \frac{1}{u} \, du - \int_1^x \frac{\{u\}}{u^2} \, du \\ &\stackrel{1}{=} 1 + \log x - \int_1^{+\infty} \frac{\{u\}}{u^2} \, du + \mathcal{O}\left(\frac{1}{x}\right) \quad (\text{as } x \rightarrow +\infty). \end{aligned}$$

Furthermore,

$$0 < \int_1^{+\infty} \frac{\{u\}}{u^2} \, du < \int_1^{+\infty} \frac{1}{u^2} \, du = 1.$$

Therefore,

$$0 < 1 - \int_1^{+\infty} \frac{\{u\}}{u^2} \, du < 1.$$

The number

$$\gamma := 1 - \int_1^{+\infty} \frac{\{u\}}{u^2} \, du$$

is the number searched for. □

¹In fact,

$$\int_x^{+\infty} \frac{\{u\}}{u^2} \, du, \frac{\{x\}}{x} \text{ are both } \mathcal{O}\left(\frac{1}{x}\right).$$

Definition 0.29. We define the following functions, known as **Chebyshev functions**:

$$\begin{aligned}\psi : (0, +\infty) &\rightarrow \mathbb{R} \\ x &\mapsto \sum_{k \leq x} \Lambda(k) = \sum_{p^m \leq x} \log p\end{aligned}$$

and

$$\begin{aligned}\vartheta : (0, +\infty) &\rightarrow \mathbb{R} \\ x &\mapsto \sum_{p \leq x} \log p.\end{aligned}$$

From the definition of these two functions, it is immediate to verify that $e^{\vartheta(x)}$ and $e^{\psi(x)}$ are, respectively, the product of all primes and the least common multiple of all positive integers less than or equal to x . Since, for every prime p , the summand $\log p$ appears exactly $\left\lfloor \frac{\log x}{\log p} \right\rfloor$ times, it is just as immediate to observe that

$$\psi(x) = \sum_{p \leq x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p.$$

The functions ψ and ϑ are related in the following way:

Proposition 0.30. *We have that*

$$\psi(x) = \sum_{n=1}^{+\infty} \vartheta(x^{\frac{1}{n}})$$

Furthermore,

$$\psi(x) = \vartheta(x) + \mathcal{O}(\sqrt{x} \log^2 x) \quad (\text{as } x \rightarrow +\infty).$$

This result about ψ and ϑ is proven in [3, Theorem 413 in Chapter XXII].

We begin with a lemma:

Lemma 0.31.

$$\sum_{n \leq x} \log n = x \log x - x + \mathcal{O}(\log x) \quad (\text{as } x \rightarrow +\infty).$$

Proof. By using Abel's lemma on summation by parts,

$$\sum_{n \leq x} \log n = [x] \log x - \int_1^x \frac{[u]}{u} du = x \log x - x + \mathcal{O}(\log x) \quad (\text{as } x \rightarrow +\infty).$$

□

The following theorems are known as Mertens' theorems:

Theorem 0.32 (Mertens).

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + \mathcal{O}(1) \quad (\text{as } x \rightarrow +\infty).$$

It is possible to prove that the error term which appears in Mertens' first theorem is

$$-\gamma + \mathcal{O}\left(\frac{1}{\log x}\right) \quad (\text{as } x \rightarrow +\infty).$$

Theorem 0.33 (Mertens). *The following holds*

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + \mathcal{O}(1) \quad (\text{as } x \rightarrow +\infty).$$

A proof of these two theorems can be found in [2, Theorem 4.3 and Theorem 4.4 in Chapter IV].

Mertens' theorems allow us to study the asymptotic behavior of the series of the reciprocals of the primes.

Corollary 0.34. *There exists a constant $C > 0$ such that*

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + C + \mathcal{O}\left(\frac{1}{\log x}\right) \quad (\text{as } x \rightarrow +\infty).$$

Proof. By using Abel's lemma on summation by parts:

$$\sum_{p \leq x} \frac{1}{p} = \sum_{p \leq x} \frac{\log p}{p} \frac{1}{\log p} = \left(\sum_{p \leq x} \frac{\log p}{p} \right) \frac{1}{\log x} + \int_2^x \left(\sum_{p \leq u} \frac{\log p}{p} \right) \frac{du}{u \log^2 u}.$$

By using Mertens' second theorem,

$$\begin{aligned} \sum_{p \leq x} \frac{1}{p} &= 1 + \mathcal{O}\left(\frac{1}{\log x}\right) + \int_2^x \frac{du}{u \log u} + \int_2^x \frac{\mathcal{O}(1)}{u \log^2 u} du \\ &= 1 + \mathcal{O}\left(\frac{1}{\log x}\right) + \int_2^x \frac{du}{u \log u} + \int_2^{+\infty} \frac{\mathcal{O}(1)}{u \log^2 u} du - \int_x^{+\infty} \frac{\mathcal{O}(1)}{u \log^2 u} du. \end{aligned}$$

Moreover, by the substitution $u = e^t$ we have

$$\int_2^x \frac{du}{u \log u} = \log \log x - \log \log 2.$$

Summing up,

$$\begin{aligned} \sum_{p \leq x} \frac{1}{p} &= \log \log x + 1 - \log \log 2 + \int_2^{+\infty} \frac{\mathcal{O}(1)}{u \log^2 u} du + \mathcal{O}\left(\frac{1}{\log x}\right) \\ &+ \mathcal{O}\left(\int_x^{+\infty} \frac{du}{u \log^2 u}\right) = \log \log x + C + \mathcal{O}\left(\frac{1}{\log x}\right) \quad (\text{as } x \rightarrow +\infty), \end{aligned}$$

where

$$C = 1 - \log \log 2 + \int_2^{+\infty} \frac{\mathcal{O}(1)}{u \log^2 u} du.$$

This completes the proof of the corollary. □

Remark 0.35. It is possible to prove that

$$C = \gamma + \sum_{p \in \mathbb{P}} \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right).$$

Prime Number Theorem

In this chapter we will provide a proof of the prime number theorem, which states that the number of primes less than or equal to x is asymptotic to $x/\log x$. We will denote by $\pi(x)$ the number of primes less than or equal to x . We present the argument in a series of steps. Specifically, we will prove a sequence of properties of the three functions

$$\zeta(s) = \sum_{n=1}^{+\infty} \frac{1}{n^s}, \quad \Phi(s) = \sum_{p \in \mathbb{P}} \frac{\log p}{p^s}, \quad \vartheta(x) = \sum_{p \leq x} \log p,$$

where s is a complex number with $\Re(s) > 1$ and x is a real number. We always use p to denote a prime. The series defining $\zeta(s)$ and $\Phi(s)$ are easily seen to be absolutely and locally uniformly convergent for $\Re(s) > 1$, so they define holomorphic functions in that domain.

1.1 Properties of $\zeta(s)$

Theorem 1.1. *We have that*

$$\zeta(s) = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{1}{p^s}} \quad \text{for } \Re(s) > 1.$$

Proof. This is a special case of Theorem 0.27. □

Theorem 1.2. $\zeta(s) - \frac{1}{s-1}$ extends holomorphically to $\Re(s) > 0$.

Proof. For $\Re(s) > 1$ we have

$$\zeta(s) - \frac{1}{s-1} = \sum_{n \in \mathbb{N}} \frac{1}{n^s} - \int_1^{+\infty} \frac{1}{x^s} dx = \sum_{n=1}^{+\infty} \int_n^{n+1} \left(\frac{1}{n^s} - \frac{1}{x^s} \right) dx.$$

The series on the right converges absolutely for $\Re(s) > 0$ because

$$\left| \int_n^{n+1} \left(\frac{1}{n^s} - \frac{1}{x^s} \right) dx \right| = \left| s \int_n^{n+1} \left(\int_n^x \frac{du}{u^{s+1}} \right) dx \right| \leq \max_{n \leq u \leq n+1} \left| \frac{s}{u^{s+1}} \right| = \frac{|s|}{n^{\Re(s)+1}},$$

by the mean value theorem. Therefore, the claim follows by applying the sandwich theorem since $\sum_n \frac{|s|}{n^{\Re(s)+1}}$ converges for $\Re(s) > 0$. $\square$

Corollary 1.3. *The zeta function ζ has a simple pole at $s = 1$.*

Proof. This follows immediately from Theorem 1.2. $\square$

Corollary 1.4. *When $s \rightarrow 1^+$, it holds that $\sum_p 1/p^s \sim \log 1/(s-1)$, and $\sum_{p,k \geq 2} 1/p^{ks}$ remains bounded.*

Proof. We have that

$$\log \zeta(s) = \sum_{p \in \mathbb{P}} -\log(1 - p^{-s}) = \sum_{p \in \mathbb{P}, k \geq 1} \frac{1}{kp^{ks}} = \sum_{p \in \mathbb{P}} 1/p^s + \psi(s),$$

with $\psi(s) = \sum_{p \in \mathbb{P}} \sum_{k \geq 2} (1/kp^{ks})$. The series ψ is majorized by the series

$$\sum_{p \in \mathbb{P}, k \geq 2} 1/p^{ks} = \sum_{p \in \mathbb{P}} 1/p^s (p^s - 1) \leq \sum_{p \in \mathbb{P}} 1/p(p-1) \leq \sum_{n=2}^{+\infty} 1/n(n-1) = 1.$$

This implies that ψ is bounded, and since $\log \zeta(s) \sim \log 1/(s-1)$, the claim follows. $\square$

The last corollary will be useful in the next chapter.

1.2 Properties of $\vartheta(x)$ and $\Phi(s)$

Proposition 1.5. $\vartheta(x) = \mathcal{O}(x)$.

Proof. First of all we recall the definition of $\vartheta(x) = \sum_{p \leq x} \log p$.

We will prove that the following estimate holds: $\vartheta(x)/x \leq 4 \log 2$. Let N be a positive integer. For every prime number p such that $N < p \leq 2N$ we have that $p \mid \binom{2N}{N}$. Indeed

$$\binom{2N}{N} = \frac{2N(2N-1) \cdots (N+1)N!}{N!N!} = \frac{2N(2N-1) \cdots (N+1)}{N!}.$$

Hence, if $p > N$ and $p \leq 2N$, p will appear in the numerator, but not in the denominator, therefore $p \mid \binom{2N}{N}$. So we get

$$2^{2N} = \sum_{k=0}^{2N} \binom{2N}{k} \geq \binom{2N}{N} \geq \prod_{\substack{p \in \mathbb{P} \\ N < p \leq 2N}} p.$$

Now we take the logarithm of the left- and right-hand side and we obtain

$$2N \log 2 \geq \log \left(\prod_{N < p \leq 2N} p \right) = \sum_{N < p \leq 2N} \log p = \vartheta(2N) - \vartheta(N).$$

Let m be a positive integer. As seen above, $\vartheta(2^m) - \vartheta(2^{m-1}) \leq 2^m \log 2$, therefore for every positive integer k we have

$$\begin{aligned} \vartheta(2^k) &= \vartheta(2^k) - \vartheta(1) = \vartheta(2^k) - \vartheta(2^{k-1}) + \vartheta(2^{k-1}) - \dots - \vartheta(2) + \vartheta(2) - \vartheta(1) \\ &\leq 2^k \log 2 + 2^{k-1} \log 2 + \dots + 2 \log 2 = \left(\sum_{h=1}^k 2^h \right) \log 2 = (2^{k+1} - 2) \log 2 \\ &= 2(2^k - 1) \log 2 < 2^{k+1} \log 2. \end{aligned}$$

Let $x \in (2, +\infty)$ and k be a positive integer such that $2^k < x \leq 2^{k+1}$. We get

$$\vartheta(x) \leq \vartheta(2^{k+1}) < 2^{k+2} \log 2 = 2^k 4 \log 2 < 4x \log 2,$$

whence

$$\frac{\vartheta(x)}{x} < 4 \log 2 \implies \limsup_{x \rightarrow +\infty} \frac{\vartheta(x)}{x} \leq 4 \log 2,$$

therefore the claim follows. $\square$

Theorem 1.6. $\zeta(s) \neq 0$ and $\Phi(s) - \frac{1}{s-1}$ is holomorphic for $\Re(s) \geq 1$.

Proof. For $\Re(s) > 1$, the convergent product in Theorem 1.1 implies that $\zeta(s) \neq 0$. By Proposition 0.19, we have $\Lambda = L * \mu$. Therefore, the Dirichlet series associated with Λ is the product of the Dirichlet series associated with¹ L and μ , that is

$$\sum_{n \in \mathbb{N}} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}.$$

By using the definition of the arithmetic function Λ , we get

$$\sum_{n \in \mathbb{N}} \frac{\Lambda(n)}{n^s} = \sum_{\substack{p \in \mathbb{P} \\ a \in \mathbb{N}}} \frac{\log p}{p^{as}} = \sum_{p \in \mathbb{P}} \log p \sum_{a \in \mathbb{N}} \left(\frac{1}{p^s} \right)^a = \sum_{p \in \mathbb{P}} \frac{\log p}{p^s - 1}.$$

¹The Dirichlet formal series associated with L is $-\zeta'(s) = \sum_{n \in \mathbb{N}} \log n / n^s$.

By some algebraic manipulations, we obtain

$$\begin{aligned} -\frac{\zeta'(s)}{\zeta(s)} &= \sum_{p \in \mathbb{P}} \frac{\log p}{p^s - 1} = \sum_{p \in \mathbb{P}} \frac{p^s \log p}{p^s(p^s - 1)} = \sum_{p \in \mathbb{P}} \frac{(p^s - 1) \log p + \log p}{p^s(p^s - 1)} \\ &= \sum_{p \in \mathbb{P}} \frac{\log p}{p^s} + \sum_{p \in \mathbb{P}} \frac{\log p}{p^s(p^s - 1)} = \Phi(s) + \sum_{p \in \mathbb{P}} \frac{\log p}{p^s(p^s - 1)}. \end{aligned}$$

The final sum converges for $\Re(s) > 1/2$, so this and Theorem 1.2 imply that $\Phi(s)$ extends meromorphically to $\Re(s) > \frac{1}{2}$, with poles only at $s = 1$ and at the zeros of $\zeta(s)$. Indeed, by definition, a pole of a function f is a zero of $1/f$, so a pole of $-\zeta'(s)/\zeta(s)$ is a zero of $-\zeta(s)/\zeta'(s)$, that is, a zero of $\zeta(s)$ or a pole of $\zeta'(s)$. Our aim is to prove that the function ζ has no zeros at s for $\Re(s) = 1$. Next, recall that if f is a meromorphic function near a point z_0 then the order of f at z_0 (which is positive if it vanishes there, negative if f has a pole there) can be computed via the residue of the logarithmic derivative of f at z_0 . In particular, for any $t \in \mathbb{R}$, the order of ζ at $1 + it$ is

$$\text{ord}_{1+it}(\zeta) = \lim_{\varepsilon \rightarrow 0^+} \frac{\varepsilon \zeta'(1 + it + \varepsilon)}{\zeta(1 + it + \varepsilon)}. \quad (1.1)$$

Moreover, recall the logarithmic derivative of ζ given by

$$-\frac{\zeta'(s)}{\zeta(s)} = \Phi(s) + \sum_{p \in \mathbb{P}} \frac{\log p}{p^s(p^s - 1)}. \quad (1.2)$$

Finally, we can remark that $\overline{\zeta(s)} = \zeta(\bar{s})$, so if ζ has a zero at $s \in \mathbb{C}$, also $\bar{s}$ is a zero of ζ . So we have that if $\zeta(s)$ has a zero of order μ at $s = 1 + i\alpha$ ($\alpha \in \mathbb{R}$, $\alpha \neq 0$) and a zero of order ν at $1 + 2i\alpha$ (so $\mu, \nu \geq 0$ by Theorem 1.2), then, by combining (1.1) and (1.2), we get

$$\lim_{\varepsilon \rightarrow 0^+} \varepsilon \Phi(1 + \varepsilon) = 1, \quad \lim_{\varepsilon \rightarrow 0^+} \varepsilon \Phi(1 + \varepsilon \pm i\alpha) = -\mu, \quad \lim_{\varepsilon \rightarrow 0^+} \varepsilon \Phi(1 + \varepsilon \pm 2i\alpha) = -\nu.$$

On the other hand, the following inequality holds:

$$\sum_{r=-2}^2 \binom{4}{2+r} \Phi(1 + \varepsilon + ir\alpha) = \sum_{p \in \mathbb{P}} \frac{\log p}{p^{1+\varepsilon}} (p^{i\alpha/2} + p^{-i\alpha/2})^4 \geq 0. \quad (1.3)$$

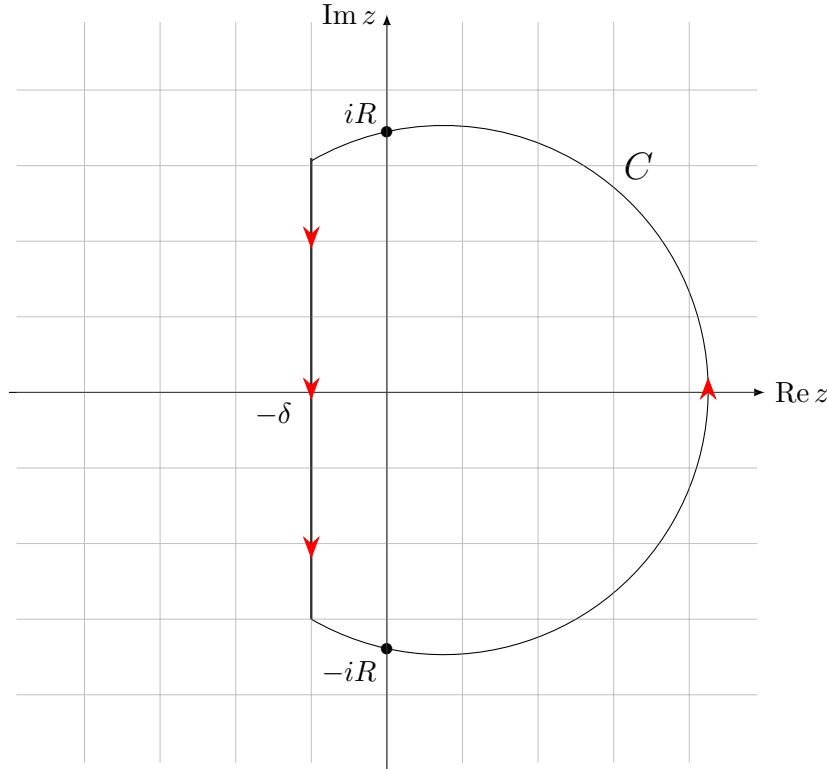
Now if we multiply left- and right-hand side of (1.3) by $\varepsilon > 0$ and then we pass to the limit as $\varepsilon \rightarrow 0^+$, we obtain that $6 - 8\mu - 2\nu \geq 0$, so $\mu = 0$, i.e., $\zeta(1 + i\alpha) \neq 0$.

□

Theorem 1.7 (Analytic theorem). *Let $f(t)$ ($t \geq 0$) be a bounded and locally integrable function and suppose that the function $g(z) = \int_0^{+\infty} f(t)e^{-zt} dt$, defined and holomorphic for $\Re(z) > 0$, extends holomorphically to $\Re(z) \geq 0$. Then $\int_0^{+\infty} f(t) dt$ exists (and equals $g(0)$).*

Proof. For $T > 0$ set $g_T(z) = \int_0^T f(t)e^{-zt} dt$. This is holomorphic for all z . We must show that $\lim_{T \rightarrow +\infty} g_T(0) = g(0)$.

Let R be large and let C be the boundary of the region $D = \{z \in \mathbb{C} \mid |z| \leq R, \Re(z) \geq -\delta\}$,



where $\delta > 0$ is small enough (depending on R) so that $g_T - g$ is holomorphic in and on C (this uses compactness of the interval $I = \{it \mid -R \leq t \leq R\}$ and holomorphicity of $g_T - g$ along I). Then

$$h(0) = g(0) - g_T(0) = \frac{1}{2\pi i} \int_C (g(z) - g_T(z)) e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z}$$

by Cauchy's theorem applied to the function $h(z) = (g(z) - g_T(z))e^{zT}(1 + z^2/R^2)$. The idea would be to estimate the contribution of the integral over C_+ , the part of C where the real part is nonnegative, and over C_- where it is nonpositive, hoping to show these contributions are small when R is large.

On the semicircle $C_+ := C \cap \{\Re(z) > 0\}$ the integrand is bounded by $\frac{2B}{R^2}$, where $B = \max_{t \geq 0} |f(t)|$, because

$$\begin{aligned} |g(z) - g_T(z)| &= \left| \int_T^{+\infty} f(t)e^{-zt} dt \right| \leq \int_T^{+\infty} |f(t)e^{-zt}| dt = \int_T^{+\infty} |f(t)| |e^{-zt}| dt \\ &\leq B \int_T^{+\infty} |e^{-zt}| dt = \frac{Be^{-\Re(z)T}}{\Re(z)} \quad (\Re(z) > 0) \end{aligned}$$

and we also have, over the circle $|z| = R$,

$$\begin{aligned} \left| e^{zT} \left(1 + \frac{z^2}{R^2} \right) \frac{1}{z} \right| &= e^{\Re(z)T} \left| \left(\frac{R^2}{z} + z \right) \frac{1}{R^2} \right| = e^{\Re(z)T} \left| \left(\frac{|z|^2}{z} + z \right) \frac{1}{R^2} \right| \\ &= e^{\Re(z)T} \left| \frac{(\bar{z} + z)}{R^2} \right| = \frac{2\Re(z)}{R^2} e^{\Re(z)T}. \end{aligned}$$

Hence the contribution to $g(0) - g_T(0)$ from the integral over C_+ is bounded in absolute value by

$$\frac{2B}{R^2} \int_{C_+} |dz| = \frac{2\pi B}{R}.$$

For the integral over $C_- := C \cap \{\Re(z) < 0\}$ we look at $g(z)$ and $g_T(z)$ separately.

Since g_T is entire, the path of integration for the integral involving g_T can be replaced by the semicircle $C'_- := \{z \in \mathbb{C} \mid |z| = R, \Re(z) < 0\}$, and the integral over C'_- is then bounded in absolute value by $\frac{2\pi B}{R}$ by exactly the same estimate as before since

$$|g_T(z)| = \left| \int_0^T f(t)e^{-zt} dt \right| \leq B \int_{-\infty}^T |e^{-zt}| dt = \frac{Be^{-\Re(z)T}}{|\Re(z)|} \quad (\Re(z) < 0).$$

Finally, the remaining integral over C_- tends to 0 as $T \rightarrow +\infty$ because the integrand is the product of the function $\frac{g(z)}{z} \left(1 + \frac{z^2}{R^2} \right)$, which is independent of T , and the function e^{zT} , which goes to 0 rapidly and uniformly on compact sets as $T \rightarrow +\infty$ in the half-plane $\Re(z) < 0$. Indeed, if we denote by $\overline{\mathcal{B}_T(0)}$ the closed disk centered at the origin with radius T , we get

$$\sup_{\substack{T \rightarrow +\infty \\ \overline{\mathcal{B}_T(0)} \cap \{\Re(z) < 0\}}} |e^{zT}| \leq \sup_{\substack{T \rightarrow +\infty \\ \overline{\mathcal{B}_T(0)} \cap \{\Re(z) < 0\}}} e^{\Re(z)T} = 0,$$

since $\Re(z) < 0$. Hence $\limsup_{T \rightarrow +\infty} |g(0) - g_T(0)| \leq \frac{4\pi B}{R}$ and, since R is arbitrary, this proves the theorem. $\square$

Proposition 1.8. *We have that*

$$\int_1^{+\infty} \frac{\vartheta(x) - x}{x^2} dx$$

is a convergent integral.

Proof. By using Abel's lemma on summation by parts, for $s > 1$ we have that

$$\sum_{p \leq x} \frac{\log p}{p^s} = \frac{1}{x^s} \sum_{p \leq x} \log p + s \int_1^x \frac{\vartheta(u)}{u^{s+1}} du.$$

Now, as $x \rightarrow +\infty$, we get

$$\Phi(s) = \lim_{x \rightarrow +\infty} \sum_{p \leq x} \frac{\log p}{p^s} = \lim_{x \rightarrow +\infty} \left(\frac{\vartheta(x)}{x^s} + s \int_1^x \frac{\vartheta(u)}{u^{s+1}} du \right) \stackrel{(1)}{=} s \int_1^{+\infty} \frac{\vartheta(u)}{u^{s+1}} du,$$

where in (1) we used Proposition 1.5 which guarantees that $\vartheta(x)/x^s$ tends to 0 as $x \rightarrow +\infty$ (for $s > 1$). By applying the substitution $u = e^t$, we obtain

$$\Phi(s) = s \int_0^{+\infty} e^{-st} \vartheta(e^t) dt.$$

Therefore, the claim follows by applying Theorem 1.7 to the two functions $f(t) = \vartheta(e^t)e^{-t} - 1$ and $g(z) = \frac{\Phi(z+1)}{z+1} - \frac{1}{z}$, which satisfy its hypotheses by Proposition 1.5 and Theorem 1.6. Indeed

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s} = \frac{1}{s+1} \left(\Phi(s+1) - \frac{1}{s} \right) - \frac{1}{s+1},$$

so $g(s)$ is holomorphic on $\Re(s) \geq 0$, since it is a sum of products of such functions. □

Proposition 1.9. $\vartheta(x) \sim x$ as $x \rightarrow +\infty$.

Proof. Assume that for some $\lambda > 1$ there are arbitrarily large x with $\vartheta(x) \geq \lambda x$. The monotonicity of ϑ implies the first inequality that follows:

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2} dt \geq \int_x^{\lambda x} \frac{\lambda x - t}{t^2} dt \stackrel{(1)}{=} \int_1^\lambda \frac{\lambda x - sx}{s^2 x} ds = \int_1^\lambda \frac{\lambda - s}{s^2} ds > 0,$$

where in (1) we used the substitution $s = t/x$. This is a contradiction, since convergence of $\int_1^{+\infty} \frac{\vartheta(t) - t}{t^2} dt$ implies that $\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2} dt$ can be made arbitrarily small by taking x large. Similarly, the inequality $\vartheta(x) \leq \lambda x$ with $\lambda < 1$ would imply

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2} dt \leq \int_x^{\lambda x} \frac{\lambda x - t}{t^2} dt \stackrel{(2)}{=} \int_1^\lambda \frac{\lambda x - tx}{t^2 x} dt = \int_1^\lambda \frac{\lambda - t}{t^2} dt < 0,$$

where in (2) the substitution is the same as the one before. There is again a contradiction for λ fixed and x big enough. □

1.3 The conclusion

Theorem 1.10 (Prime number theorem). $\pi(x) \sim \frac{x}{\log x}$ as $x \rightarrow +\infty$, that is, $\lim_{x \rightarrow +\infty} (\pi(x) \log x) / x = 1$.

Proof. The prime number theorem follows easily from Proposition 1.9, since for any $\varepsilon > 0$

$$\vartheta(x) = \sum_{p \leq x} \log p \leq \sum_{p \leq x} \log x = \pi(x) \log x, \quad (1.4)$$

$$\begin{aligned} \vartheta(x) &\geq \sum_{x^{1-\varepsilon} \leq p \leq x} \log p \geq \sum_{x^{1-\varepsilon} \leq p \leq x} (1 - \varepsilon) \log x \\ &= (1 - \varepsilon) \log x \left(\sum_{x^{1-\varepsilon} \leq p \leq x} 1 \right) = (1 - \varepsilon) \log x \left(\pi(x) - \pi(x^{1-\varepsilon}) \right) \\ &= (1 - \varepsilon) \log x \left(\pi(x) + \mathcal{O}(x^{1-\varepsilon}) \right). \end{aligned} \quad (1.5)$$

By combining equations (1.4) and (1.5), we obtain

$$(1 - \varepsilon) \log x \left(\pi(x) + \mathcal{O}(x^{1-\varepsilon}) \right) \leq \vartheta(x) \leq \pi(x) \log x.$$

Now we divide each member by x , so we have

$$(1 - \varepsilon) \left(\frac{\pi(x)}{(x/\log x)} + \mathcal{O}(x^{1-\varepsilon}) \right) \leq \frac{\vartheta(x)}{x} \leq \frac{\pi(x)}{(x/\log x)}.$$

By applying the result of Proposition 1.9, i.e., $\vartheta(x) = x + o(x)$, for any ε we get

$$(1 - \varepsilon) \left(\frac{\pi(x)}{(x/\log x)} + \mathcal{O}(x^{1-\varepsilon}) \right) \leq 1 + o(1) \leq \frac{\pi(x)}{(x/\log x)}.$$

In conclusion, as $x \rightarrow +\infty$, due to arbitrariness of ε we obtain $\pi(x) \sim \frac{x}{\log x}$ and the claim follows. □

Equivalence between logarithmic and analytic density

In this chapter we want to prove the equivalence between logarithmic and analytic density and, subsequently, we are going to describe an example of a subset of prime numbers that admits logarithmic density but not natural density.

Definition 2.1. Let S be a set consisting of prime numbers. We will call **natural density** of S the quantity

$$\text{dens}_N(S) = \lim_{n \rightarrow +\infty} \frac{|S \cap \{1, \dots, n\}|}{\pi(n)},$$

if this limit exists.

Definition 2.2. Let S be a set consisting of prime numbers. We will call **logarithmic density** of S the quantity

$$\text{dens}_L(S) = \lim_{x \rightarrow +\infty} \frac{\sum_{p \in S, p \leq x} 1/p}{\sum_{p \leq x} 1/p},$$

if this limit exists.

Definition 2.3. Let S be a set consisting of prime numbers. We will call **Dirichlet density** (or **analytic density**) of S the quantity

$$\text{dens}_A(S) = \lim_{s \rightarrow 1^+} \frac{\sum_{p \in S} 1/p^s}{\sum_{p \in \mathbb{P}} 1/p^s},$$

if this limit exists.

Remark 2.4. For $\bullet = N, L, A$ the quantity $\text{dens}_\bullet(S)$, if it exists, is a real number between 0 and 1.

Let us remark that $\sum_{p \leq x} 1/p \sim \log \log x$ by Corollary 0.34 and $\sum_{p \in \mathbb{P}} 1/p^s \sim \log(1/(s-1))$ thanks to Corollary 1.4.

Theorem 2.5. *Let S be a set consisting of prime numbers. S admits analytic density if and only if it admits logarithmic density, and in that case $\text{dens}_A(S) = \text{dens}_L(S)$.*

We will prove this theorem in the next two sections.

2.1 From logarithmic density to analytic density

Let S be a set consisting of prime numbers and let

$$a_n = \begin{cases} 1/n & \text{if } n \in S, \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, set $A(u) = \sum_{n \leq u} a_n$, $\phi(u) = u^{1-s}$, with $s > 1$. Let us assume that S admits logarithmic density, that is, $A(u) = \sum_{p \in S, p \leq u} 1/p$ satisfies the following relation

$$A(u) = (\delta_S + o(1)) \log \log(u),$$

for some $\delta_S \in [0, 1]$. By applying Abel's lemma on summation by parts on a_n and $\phi(u)$, we obtain

$$\sum_{n \leq t} a_n \phi(n) = \left(\sum_{p \in S, p \leq t} \frac{1}{p} \right) t^{1-s} + (s-1) \int_1^t A(u) u^{-s} du.$$

We have that $\sum_{p \in S, p \leq t} 1/p \leq \sum_{p \in \mathbb{P}, p \leq t} 1/p \sim \log \log t$. Since $s > 1$, the factor t^{1-s} decreases much faster than that $\log \log t$ increases. Therefore, the quantity $(\sum_{p \in S, p \leq t} 1/p)/t^{s-1}$ tends to 0 for $t \rightarrow +\infty$. So, by taking the limit for $t \rightarrow +\infty$, we get

$$\sum_{p \in S} \frac{1}{p^s} = (s-1) \int_1^{+\infty} A(u) u^{-s} du.$$

Since $A(u) = (\delta_S + o(1)) \log \log(u)$ by assumption, it is enough to prove the following proposition

Proposition 2.6. *The following estimate holds*

$$(s-1) \int_1^{+\infty} u^{-s} \log \log(u) du \sim \log \left(\frac{1}{s-1} \right) \quad (\text{as } s \rightarrow 1^+).$$

Proof. Set

$$F(s) := (s-1) \int_1^{+\infty} u^{-s} \log \log(u) du = (s-1) \int_0^{+\infty} e^{t(1-s)} \log(t) dt.$$

By deriving under the integral sign,

$$F'(s) = \int_0^{+\infty} e^{t(1-s)} \log(t) dt - (s-1) \int_0^{+\infty} e^{t(1-s)} t \log(t) dt.$$

Now we can integrate by parts to obtain

$$\begin{aligned} & (s-1) \int_0^{+\infty} e^{t(1-s)} t \log(t) dt \\ &= (s-1) \left(\left[t \log(t) \frac{e^{t(1-s)}}{1-s} \right]_0^{+\infty} - \int_0^{+\infty} (1 + \log(t)) \frac{e^{t(1-s)}}{1-s} dt \right) \\ &= \int_0^{+\infty} (1 + \log(t)) e^{t(1-s)} dt, \end{aligned}$$

therefore

$$\begin{aligned} F'(s) &= \int_0^{+\infty} \log(t) e^{t(1-s)} dt - \int_0^{+\infty} (1 + \log(t)) e^{t(1-s)} dt \\ &= - \int_0^{+\infty} e^{t(1-s)} dt = - \left[\frac{e^{t(1-s)}}{1-s} \right]_0^{+\infty} = \frac{1}{1-s}. \end{aligned}$$

So we can conclude that

$$F(s) = -\log(s-1) + C_1 = \log\left(\frac{1}{s-1}\right) + C_1.$$

It is possible to determine the constant C_1 , for example if we set $s = 2$, we get

$$F(2) = \int_1^{+\infty} u^{-2} \log \log(u) du = \log\left(\frac{1}{2-1}\right) + C_1 = C_1.$$

Now if we apply the substitution $u = 1/x$ in (1) and $x = e^t$ in (2), we obtain

$$\begin{aligned} C_1 &= \int_1^{+\infty} u^{-2} \log \log(u) du \stackrel{(1)}{=} \int_0^1 \log \log(1/x) dx \\ &\stackrel{(2)}{=} \int_0^{+\infty} e^{-t} \log t dt = \Gamma'(1) = -\gamma, \end{aligned}$$

where γ is the Euler-Mascheroni constant and $\Gamma(z)$ is the Gamma function. In conclusion, the claim follows. $\square$

2.2 From analytic density to logarithmic density

We need two estimates on prime numbers.

Proposition 2.7. *The following holds*

$$\sum_{p \leq t} \left(\frac{1}{p} - \frac{1}{p^{1+1/\log t}} \right) = \mathcal{O}(1),$$

in particular it is $o(\log \log t)$.

Proof. Recall the estimate obtained by Mertens' theorems in Corollary 0.34, that is, $\sum_{p \leq x} 1/p = \log \log x + C + \mathcal{O}(1/\log x)$.

By difference, we have that for $x > y$

$$\sum_{y < p \leq x} \frac{1}{p} = \log \log x - \log \log y + \mathcal{O}(1/\log y).$$

Now we can rewrite the sum that interests us in the following way

$$\sum_{p \leq t} \frac{e^{\log p / \log t} - 1}{p^{1+1/\log t}}$$

and we can note that $e^{\log p / \log t} - 1 \ll \log p / \log t$ for $p \leq t$. Let $n_{\max}$ be the greatest integer such that $t^{1/n} \geq 2$ (therefore $n_{\max} \ll \log t$). So we have

$$\begin{aligned} \sum_{p \leq t} \frac{e^{\log p / \log t} - 1}{p^{1+1/\log t}} &\ll \sum_{p \leq t} \frac{\log p / \log t}{p} \\ &= \sum_{n=1}^{n_{\max}} \sum_{t^{1/(n+1)} < p \leq t^{1/n}} \frac{\log p / \log t}{p} \leq \sum_{n=1}^{n_{\max}} \frac{1}{n} \sum_{t^{1/(n+1)} < p \leq t^{1/n}} \frac{1}{p} \\ &\leq \sum_{n=1}^{n_{\max}} \frac{1}{n} \left(\log \log t^{1/n} - \log \log t^{1/(n+1)} + \mathcal{O}(1/\log(t^{1/(n+1)})) \right) \\ &\leq \sum_{n=1}^{n_{\max}} \frac{1}{n} \left(\log \left(\frac{1}{n} \log t \right) - \log \left(\frac{1}{n+1} \log t \right) + \mathcal{O}\left(\frac{n}{\log t}\right) \right) \\ &= \sum_{n=1}^{n_{\max}} \frac{1}{n} \left(\log \left(\frac{n+1}{n} \right) + \mathcal{O}\left(\frac{n}{\log t}\right) \right) \\ &\leq \sum_{n=1}^{n_{\max}} \frac{1}{n^2} + \sum_{n=1}^{n_{\max}} \mathcal{O}\left(\frac{1}{\log t}\right) \\ &\leq \zeta(2) + \mathcal{O}\left(\frac{n_{\max}}{\log t}\right) = \mathcal{O}(1). \end{aligned}$$

The claim follows. □

Proposition 2.8. *It holds that*

$$\sum_{p>t} \frac{1}{p^{1+1/\log t}} = \mathcal{O}(1),$$

in particular it is $o(\log \log t)$.

Proof. First of all, we can replace the condition $p > t$ with $p > t \log \log t$: indeed, the difference

$$\sum_{p>t} \frac{1}{p^{1+1/\log t}} - \sum_{p>t \log \log t} \frac{1}{p^{1+1/\log t}} = \sum_{t < p \leq t \log \log t} 1/p^{1+1/\log t}$$

is smaller than the sum

$$\begin{aligned} \sum_{t < p \leq t \log \log t} \frac{1}{p} &= \log \log(t \log \log t) - \log \log(t) + \mathcal{O}(1/\log t) \\ &= \log \left(\frac{\log(t \log \log t)}{\log t} \right) + \mathcal{O}\left(\frac{1}{\log t}\right) \\ &= \log \left(\frac{\log t + \log \log \log t}{\log t} \right) + \mathcal{O}\left(\frac{1}{\log t}\right) \\ &= \log \left(1 + \frac{\log \log \log t}{\log t} \right) + \mathcal{O}\left(\frac{1}{\log t}\right) \\ &\leq \frac{\log \log \log t}{\log t} + \mathcal{O}\left(\frac{1}{\log t}\right) = \mathcal{O}(1). \end{aligned}$$

Let $B := t \log \log t$ and let us focus on the sum

$$\begin{aligned} \sum_{p>B} p^{-1-1/\log t} &= \sum_{k=1}^{+\infty} \sum_{kB < p \leq (k+1)B} p^{-1-1/\log t} \\ &\leq \sum_{k \geq 1} (kB)^{-1-1/\log t} \cdot |\{p \in \mathbb{P} \mid kB < p \leq (k+1)B\}|. \end{aligned}$$

Let us call $A_k = (kB)^{-1-1/\log t}$ and $b_k = |\{p \in \mathbb{P} \mid kB < p \leq (k+1)B\}|$. So the previous sum is

$$\begin{aligned} \sum_{k=1}^n A_k b_k &= A_n \sum_{k=1}^n b_k - \sum_{k=1}^{n-1} \left((A_{k+1} - A_k) \sum_{j=1}^k b_j \right) \\ &= A_n \sum_{k=1}^n b_k + \sum_{k=1}^{n-1} \left((A_k - A_{k+1}) \sum_{j=1}^k b_j \right). \end{aligned}$$

Let us observe that $\sum_{j=1}^n b_j = \pi((n+1)B) - \pi(B) \leq \pi((n+1)B) \ll \frac{(n+1)B}{\log((n+1)B)}$, thanks to a weak version of the prime number theorem. In particular, the first term above, given by

$$A_n \sum_{k=1}^n b_k \ll (nB)^{-1-1/\log t} \frac{(n+1)B}{\log((n+1)B)} \ll \frac{1}{\log(n+1)},$$

is infinitesimal in n . Now we can pass to the limit as $n \rightarrow +\infty$ and obtain

$$\sum_{p>B} p^{-1-1/\log t} = \sum_{k \geq 1} A_k b_k = \sum_{k=1}^{+\infty} \left((A_k - A_{k+1}) \sum_{j=1}^k b_j \right). \quad (2.1)$$

We now remark that

$$\begin{aligned} A_k - A_{k+1} &= \frac{1}{B^{1+1/\log t}} \left(\frac{1}{k^{1+1/\log t}} - \frac{1}{(k+1)^{1+1/\log t}} \right) \\ &= \frac{1}{B^{1+1/\log t}} \cdot \frac{1}{(k+1)^{1+1/\log t}} \cdot \left(\left(\frac{k+1}{k} \right)^{1+1/\log t} - 1 \right) \\ &\ll \frac{1}{B^{1+1/\log t}} \cdot \frac{1}{(k+1)^{1+1/\log t}} \cdot \frac{1}{k}. \end{aligned}$$

By replacing in equation (2.1), we get

$$\begin{aligned} \sum_{p>B} p^{-1-1/\log t} &= \sum_{k=1}^{+\infty} \left((A_k - A_{k+1}) \sum_{j=1}^k b_j \right) \\ &\ll \sum_{k=1}^{+\infty} \frac{1}{B^{1+1/\log t}} \cdot \frac{1}{(k+1)^{1+1/\log t}} \cdot \frac{1}{k} \cdot \frac{(k+1)B}{\log((k+1)B)} \\ &\ll \sum_{k=1}^{+\infty} \frac{1}{B^{1/\log t}} \cdot \frac{1}{(k+1)^{1+1/\log t}} \cdot \frac{1}{\log((k+1)B)} \\ &\ll \frac{1}{\log(B)B^{1/\log t}} \sum_{k=1}^{+\infty} \frac{1}{(k+1)^{1+1/\log t}} \\ &\ll \frac{1}{\log B} \sum_{k=1}^{+\infty} \frac{1}{(k+1)^{1+1/\log t}} \leq \frac{1}{\log B} \zeta(1+1/\log t), \end{aligned}$$

where in the last line we have used the following relation

$$B^{1/\log t} = \exp\left(\frac{\log B}{\log t}\right) = \exp\left(\frac{\log t + \log \log \log t}{\log t}\right) = e + o(1).$$

Now, by remarking that $\zeta(1+s) \sim 1/s$ for $s \rightarrow 0^+$, we obtain that the whole sum is $\ll \frac{1}{\log B} \log t = \mathcal{O}(1)$. □

Let S be a set consisting of prime numbers and let us assume that S admits analytic density, that is, the function $N(s) := \frac{\sum_{p \in S} 1/p^s}{\sum_{p \in \mathbb{P}} 1/p^s}$ satisfies the asymptotic estimate $N(s) = (\delta_S + o(1)) \log(1/(s-1))$ for $s \rightarrow 1^+$, where $\delta_S \in [0, 1]$. By using Propositions 2.7 and 2.8, we have

$$\begin{aligned} \sum_{p \in S, p \leq t} \frac{1}{p} &= \sum_{p \in S, p \leq t} \left(\frac{1}{p} - \frac{1}{p^{1+1/\log t}} \right) + \sum_{p \in S, p \leq t} \frac{1}{p^{1+1/\log t}} \\ &= o(\log \log t) + \sum_{p \in S, p \leq t} \frac{1}{p^{1+1/\log t}} \stackrel{(1)}{=} \sum_{p \in S} \frac{1}{p^{1+1/\log t}} + o(\log \log t) \\ &= N(1 + 1/\log t) + o(\log \log t) = (\delta_S + o(1)) \log \log t + o(\log \log t), \end{aligned}$$

where in (1) we used the fact that adding the primes $> t$ to the sum does not change the result since they are $\mathcal{O}(1)$. If we divide by $\log \log t$ and, then, we pass to the limit, the claim follows.

2.3 A singular subset of $\mathbb{P}$

Now we want to prove that if a set consisting of prime numbers admits natural density, then it also admits logarithmic density (or, equivalently, analytic density), but the other implication does not hold, that is, there exist subsets of $\mathbb{P}$ which admit logarithmic density but not natural density.

Proposition 2.9. *Let f be an arithmetic function and S be a set consisting of prime numbers. If the limit*

$$\lim_{x \rightarrow +\infty} \frac{1}{\pi(x)} \sum_{p \in S, p \leq x} f(p)$$

exists, then so does

$$\lim_{x \rightarrow +\infty} \frac{1}{\log \log x} \sum_{p \in S, p \leq x} \frac{f(p)}{p},$$

and their values agree.

Proof. We can rewrite the hypothesis in the following way:

$$\sum_{p \in S, p \leq x} f(p) = \lambda \pi(x) + o(\pi(x)) = \frac{\lambda x}{\log x} + o\left(\frac{x}{\log x}\right),$$

for some $\lambda \in \mathbb{R}$. By applying Abel's lemma on summation by parts, we obtain

$$\begin{aligned} \sum_{p \in S, p \leq x} \frac{f(p)}{p} &\stackrel{(1)}{=} \left(\sum_{p \in S, p \leq x} f(p) \right) \frac{1}{x} + \int_2^x \frac{1}{u^2} \sum_{p \in S, p \leq u} f(p) du \\ &= \frac{1}{x} \frac{\lambda x}{\log x} + o\left(\frac{1}{\log x}\right) + \int_2^x \frac{\lambda u}{u^2 \log u} du + o\left(\int_2^x \frac{u}{u^2 \log u} du\right) \\ &= \frac{\lambda}{\log x} + \lambda \log \log x - \lambda \log \log 2 + o(\log \log x) \\ &\stackrel{(2)}{=} \lambda \log \log x + o(\log \log x). \end{aligned}$$

In (1) we made the integral start from 2 to simplify the computation since S is a set of primes and in (2) we incorporated $1/\log x$ and $\log \log 2$ into the error term. Now we divide by $\log \log x$ and we get

$$\frac{1}{\log \log x} \sum_{p \in S, p \leq x} \frac{f(p)}{p} = \lambda + o(1).$$

In conclusion, we pass to the limit as $x \rightarrow +\infty$ and the claim follows. □

Corollary 2.10. *Let S be a set consisting of prime numbers. If S admits natural density, then it also admits logarithmic density.*

Proof. Let f be the characteristic function of the set S , that is

$$f(p) = \begin{cases} 1 & \text{if } p \in S, \\ 0 & \text{otherwise.} \end{cases}$$

By applying Proposition 2.9 to S and f , we get that S admits logarithmic density and its value coincides with the natural density

$$\text{dens}_N(S) = \lim_{n \rightarrow +\infty} \frac{|S \cap \{1, \dots, n\}|}{\pi(n)} = \lim_{x \rightarrow +\infty} \frac{1}{\pi(x)} \sum_{p \in S, p \leq x} 1,$$

that exists by hypothesis. Hence the claim follows. □

We now provide a counterexample to the opposite implication of Corollary 2.10.

Let $M := \{p \in \mathbb{P} \mid p \text{ starts with 1 in the decimal representation}\}$.

Lemma 2.11. *The set M admits logarithmic density, equal to $\frac{\log 2}{\log 10}$, but not natural density.*

We will prove Lemma 2.11 in the following propositions.

Proposition 2.12. *M does not admit natural density.*

Proof. Let $M(x) = |\{p \in M \mid p \leq x\}|$. Immediately we can give these two inequalities:

- i) $M(10^k) \leq \pi(2 \cdot 10^{k-1})$, since there are no primes that begin with 1 between $2 \cdot 10^{k-1}$ and 10^k .
- ii) $M(2 \cdot 10^k) \geq \pi(2 \cdot 10^k) - \pi(10^k)$, since $M(2 \cdot 10^k)$ counts the primes smaller than $2 \cdot 10^k$ that start with 1 in the decimal representation, and we are excluding all prime numbers smaller than 10^k . Therefore, the expression $\pi(2 \cdot 10^k) - \pi(10^k)$ considers only the primes between 10^k and $2 \cdot 10^k$, that, clearly, begin with 1 in decimal representation.

By applying the prime number theorem, we obtain the following relation

$$\frac{M(10^k)}{\pi(10^k)} \leq \frac{\pi(2 \cdot 10^{k-1})}{\pi(10^k)} \sim \frac{2 \cdot 10^{k-1}}{\log(2 \cdot 10^{k-1})} \cdot \frac{\log(10^k)}{10^k} = \frac{1}{5} \frac{\log(10^k)}{\log(2 \cdot 10^{k-1})}.$$

Now we operate some algebraic manipulations on the last expression as follows:

$$\frac{1}{5} \cdot \frac{\log(10^k)}{\log(2 \cdot 10^{k-1})} = \frac{1}{5} \cdot \frac{k \log 10}{\log 2 + (k-1) \log 10} = \frac{1}{5} \left(1 + \frac{\log 5}{\log(2 \cdot 10^{k-1})} \right).$$

Therefore, we have

$$\frac{M(10^k)}{\pi(10^k)} \leq \frac{\pi(2 \cdot 10^{k-1})}{\pi(10^k)} \sim \frac{1}{5} \left(1 + \frac{\log 5}{\log(2 \cdot 10^{k-1})} \right).$$

If we pass to the limit as $k \rightarrow +\infty$, the term on the right tends to $1/5$.

On the other hand, from the relation (ii) we have

$$\frac{M(2 \cdot 10^k)}{\pi(2 \cdot 10^k)} \geq 1 - \frac{\pi(10^k)}{\pi(2 \cdot 10^k)} \sim 1 - \frac{10^k}{\log(10^k)} \cdot \frac{\log(2 \cdot 10^k)}{2 \cdot 10^k} = 1 - \frac{1}{2} \left(1 + \frac{\log 2}{k \log 10} \right).$$

The term on the right tends to $1/2$ as $k \rightarrow +\infty$. So we have that the value $\frac{M(x)}{\pi(x)}$ is frequently greater than or equal to $\frac{1}{2} - \varepsilon$, and frequently less than or equal to $\frac{1}{5} + \varepsilon$ for every $\varepsilon > 0^-$, consequently, it does not admit limit as $x \rightarrow +\infty$.

□

Proposition 2.13. *M admits logarithmic density.*

Proof. We want to compute the following limit:

$$\text{dens}_L(M) = \lim_{x \rightarrow +\infty} \frac{\sum_{p \in S, p \leq x} 1/p}{\sum_{p \leq x} 1/p}.$$

By using Mertens' theorems, we can write

$$\sum_{x \leq p < 2x} \frac{1}{p} = \log \log 2x - \log \log x + \mathcal{O}\left(\frac{1}{(\log x)^2}\right),$$

where the error term is different from the one in the Corollary 0.34. For the $\mathcal{O}(1/(\log x)^2)$ error term, we need a sufficiently good error term in the prime number theorem, then a summation by parts yields the given (or a stronger) error term straightforwardly. For a proof of this fact, see [5]. Let us suppose that k is a positive integer such that $2 \cdot 10^k \leq x < 2 \cdot 10^{k+1}$. Then we get

$$\sum_{p \in S, p \leq x} \frac{1}{p} = \sum_{h=1}^k \left(\sum_{10^h \leq p < 2 \cdot 10^h} \frac{1}{p} \right) \quad (2.2)$$

Let us focus on the term inside the brackets. By applying the previous remark, we obtain

$$\begin{aligned} \sum_{10^h \leq p < 2 \cdot 10^h} \frac{1}{p} &= \log \log(2 \cdot 10^h) - \log \log(10^h) + R(h) \\ &= \log \left(1 + \frac{\log 2}{h \log 10} \right) + R(h) \sim \frac{1}{h} \cdot \frac{\log 2}{\log 10} + R(h), \end{aligned}$$

where the error term $R(h)$ is $\mathcal{O}(h^{-2})$. Now we denote by $A = \frac{\log 2}{\log 10}$ and apply Theorem 0.28, so equation (2.2) assumes the form

$$\sum_{p \in S, p \leq x} \frac{1}{p} = \sum_{h=1}^k \left(\frac{A}{h} + R(h) \right) = A \sum_{h=1}^k \frac{1}{h} + \sum_{h=1}^k R(h) = A \log k + A\gamma + \mathcal{O}(k^{-1}) + C,$$

with $C = \sum_{h=1}^{+\infty} R(h)$. Set $k = \left\lfloor \frac{\log(x/2)}{\log 10} \right\rfloor = \left(\frac{\log x - \log 2}{\log 10} \right) \left(1 + \mathcal{O}\left(\frac{1}{\log x}\right) \right)$.

Therefore we obtain

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{\sum_{p \in S, p \leq x} 1/p}{\sum_{p \leq x} 1/p} &= \lim_{x \rightarrow +\infty} \frac{A \log \left(\frac{\log x - \log 2}{\log 10} \right) + A\gamma + C + \mathcal{O}(1/\log x)}{\log \log x} \\ &= \lim_{x \rightarrow +\infty} \frac{A \log(\log x - \log 2) - A \log \log 10 + A\gamma + C + \mathcal{O}(1/\log x)}{\log \log x} = A. \end{aligned}$$

Hence, the claim follows. □

Dirichlet's theorem

The aim of this chapter is to prove the following theorem, conjectured (and used) by Legendre and proved by Dirichlet.

Theorem 3.1 (Dirichlet's theorem). *Let a and m be relatively prime integers greater than or equal to 1. There exist infinitely many prime numbers p such that $p \equiv a \pmod{m}$.*

The method we follow (which is that of Dirichlet himself) uses the properties of *L-functions*.

3.1 Characters of finite abelian groups

3.1.1 Duality

Let G be a finite abelian group written multiplicatively.

Definition 3.2. A **character** of G is a homomorphism of G into the multiplicative group $\mathbb{C}^*$ of nonzero complex numbers.

The characters of G form a group $\text{Hom}(G, \mathbb{C}^*)$ which we will denote by $\hat{G}$, called the **dual** of G .

Example 3.3. Suppose that G is cyclic of order n with generator s . If $\chi : G \rightarrow \mathbb{C}^*$ is a character of G , the element $w = \chi(s)$ satisfies the relation $w^n = 1$, i.e., it is an n -th root of unity. Conversely every n -th root of unity w defines a character of G by means of $s^a \mapsto w^a$. Thus we see that the map $\chi \mapsto \chi(s)$ is an isomorphism of $\hat{G}$ on the group μ_n of n -th roots of unity. In particular, $\hat{G}$ is cyclic of order n .

Proposition 3.4. *Let H be a subgroup of G . Every character of H extends to a character of G .*

Proof. We use induction on the index $[G : H]$ of H in G .

If $[G : H] = 1$, then $H = G$ and there is nothing to prove.

Otherwise, let x be an element of G not contained in H , and let n be the smallest integer greater than 1 such that $x^n \in H$ (in the worst case n is equal to the order of x). Let χ be a character of H , and let $t = \chi(x^n)$. Since $\mathbb{C}^*$ is a *divisible group*¹, one can choose an element $w \in \mathbb{C}^*$ such that $w^n = t$. Let H' be the subgroup of G generated by H and x ; every element h' of H' can be written in the form $h' = hx^a$ with $a \in \mathbb{Z}$ and $h \in H$. Put

$$\chi'(h') = \chi(h)w^a.$$

It is possible to prove that this number does not depend on the decomposition hx^a of h' and that $\chi' : H' \rightarrow \mathbb{C}^*$ is a character of H' extending χ .

Since we have $[G : H'] < [G : H]$ the inductive hypothesis allows us to extend χ' to a character of G . □

Remark 3.5. The operation of restriction defines a homomorphism

$$\rho : \hat{G} \rightarrow \hat{H}$$

and Proposition 3.4 says that ρ is surjective. Moreover, the kernel of ρ is the set of characters of G which are trivial on H ; it is thus isomorphic to the group $(\widehat{G/H})$ dual to G/H . Hence the exact sequence:

$$\{1\} \rightarrow (\widehat{G/H}) \rightarrow \hat{G} \rightarrow \hat{H} \rightarrow \{1\}.$$

Proposition 3.6. *The group $\hat{G}$ is a finite abelian group of the same order as G .*

Proof. We can use induction on the order n of G .

The case $n = 1$ is trivial. If $n \geq 2$, we can choose a non-trivial cyclic subgroup H of G . By the remark above, the order of $\hat{G}$ is the product of the orders of $\hat{H}$ and of $(\widehat{G/H})$. But the order of H (respectively of G/H) is equal to that of its dual, because H is cyclic (respectively because G/H is of order strictly smaller than n). We conclude from this that the order of $\hat{G}$ is the product of the orders of H and of G/H , hence is equal to the order of G . □

¹a **divisible group** is an abelian group in which every element can, in some sense, be divided by positive integers, or more accurately, every element is an n -th multiple for each positive integer n . In formulas, an abelian group $(G, \cdot)$ is *divisible* if, for every positive integer n and every $g \in G$, there exists $y \in G$ such that $y^n = g$.

Remark 3.7. It is possible to prove a more precise result: $\hat{G}$ is isomorphic (non-canonically in general) to G . This is shown by decomposing G into a product of cyclic groups.

If $x \in G$ the function $\chi \mapsto \chi(x)$ is a character of $\hat{G}$. We obtain thus a homomorphism $\varepsilon : G \rightarrow \hat{\hat{G}}$.

Proposition 3.8. *The homomorphism ε is an isomorphism of G onto its bidual $\hat{\hat{G}}$.*

Proof. Since G and $\hat{\hat{G}}$ have the same order, it suffices to prove that ε is injective, i.e. that, if x is an element of G and it is not equal to 1, there exists a character χ of G such that $\chi(x) \neq 1$. Now, let H be the cyclic subgroup of G generated by x . It is clear (see Example 3.3) that there exists a character χ of H such that $\chi(x) \neq 1$ and Proposition 3.4 shows that χ extends to a character of G ; hence the desired result. $\square$

3.1.2 Orthogonality relations

Proposition 3.9. *Let $n = |G|$ and let $\chi \in \hat{G}$. Then*

$$\sum_{x \in G} \chi(x) = \begin{cases} n & \text{if } \chi = 1, \\ 0 & \text{if } \chi \neq 1. \end{cases}$$

Proof. If $\chi = 1$, then $\chi(x) = 1$ for every $x \in G$, therefore the first formula is obvious. To prove the second one, since $\chi \neq 1$ we can choose $y \in G$ such that $\chi(y) \neq 1$. So we have

$$\chi(y) \sum_{x \in G} \chi(x) = \sum_{x \in G} \chi(xy) = \sum_{x \in G} \chi(x),$$

hence

$$(\chi(y) - 1) \sum_{x \in G} \chi(x) = 0.$$

Since $\chi(y) \neq 1$, this implies $\sum_{x \in G} \chi(x) = 0$. $\square$

Corollary 3.10. *Let $x \in G$. Then*

$$\sum_{\chi \in \hat{G}} \chi(x) = \begin{cases} n & \text{if } x = 1, \\ 0 & \text{if } x \neq 1. \end{cases}$$

Proof. This follows from Proposition 3.9 applied to the dual group $\hat{G}$ and the identification of $\psi \in \hat{\hat{G}}$ with “evaluating in $x \in G$ ”, that is, Proposition 3.8. $\square$

3.1.3 Modular characters

Let m be an integer greater than or equal to 1. We denote by $G(m)$ the multiplicative group $(\mathbb{Z}/m\mathbb{Z})^*$ of invertible elements of the ring $(\mathbb{Z}/m\mathbb{Z})$. It is an abelian group of order $\varphi(m)$, where φ is Euler's function of m .

Definition 3.11. An element χ of the dual of $G(m)$ is called a **character modulo m** .

It can be viewed as a function, defined on the set of integers prime to m , with values in $\mathbb{C}^*$, and such that $\chi(ab) = \chi(a)\chi(b)$; it is convenient to extend such a function to all of $\mathbb{Z}$ by putting $\chi(a) = 0$ if a is not prime to m .

Example 3.12. 1) $m = p$ with p prime not equal to 2. The group $G(p)$ is cyclic of order $p - 1$, hence has a unique character of order 2, the *Legendre character* $x \mapsto \left(\frac{x}{p}\right)$.

2) $m = 7$. The group $G(7)$ is cyclic of order 6, thus has two characters of order 3 which are complex conjugates. One of them is given by

$$\begin{aligned}\chi(x) &= 1 && \text{if } x \equiv \pm 1 \pmod{7}, \\ \chi(x) &= e^{2\pi i/3} && \text{if } x \equiv \pm 2 \pmod{7}, \\ \chi(x) &= e^{4\pi i/3} && \text{if } x \equiv \pm 4 \pmod{7}.\end{aligned}$$

3.2 Review of complex analysis

Lemma 3.13. *Let U be an open subset of $\mathbb{C}$ and let f_n be a sequence of holomorphic functions on U which converges uniformly on every compact set to a function f . Then f is holomorphic in U and the derivatives f'_n of the f_n converge uniformly on all compact subsets to the derivative f' of f .*

Proof. To show that f is holomorphic it is sufficient, by Morera's theorem, to show that the differential form $f(z)dz$ is closed. For this, it suffices to show that for every D , closed disc contained in U , $\int_\gamma f(z)dz = 0$, where γ is the boundary of D , oriented in the usual manner. Since f is the uniform limit of a sequence of holomorphic functions f_n , we have that

$$\int_\gamma f(z)dz = \lim_{n \rightarrow +\infty} \int_\gamma f_n(z)dz = 0.$$

So, the first part of the lemma follows. For the second part:

Let $K \subset U$ be compact and suppose $a \in K$. Since U is open, for some $\varepsilon > 0$ the compact set $K_\varepsilon = K \cup \overline{\mathcal{B}_\varepsilon(a)}$ is contained in U . Set $\gamma = \partial \overline{\mathcal{B}_\varepsilon(a)}$, i.e., the boundary of $\mathcal{B}_\varepsilon(a)$, oriented in the usual manner. Then we have by Cauchy's

integral formula and by differentiating with respect to z under the integral sign

$$f'(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^2} dz = \frac{1}{2\pi i} \lim_{n \rightarrow +\infty} \int_{\gamma} \frac{f_n(z)}{(z-a)^2} dz = \lim_{n \rightarrow +\infty} f'_n(a).$$

Since $f_n \rightarrow f$ uniformly on K_ε , we have that

$$|f'(a) - f'_n(a)| \leq \frac{1}{2\pi} \frac{l(\gamma)}{\varepsilon^2} \sup_{z \in K_\varepsilon} |f(z) - f_n(z)|,$$

where $l(\gamma)$ denotes the length of γ . So, we see that $f'_n \rightarrow f'$ uniformly on K_ε , hence on K and the lemma follows. $\square$

Lemma 3.14. *Let α, β be two real numbers with $0 < \alpha < \beta$. Let $z = x + iy$ with $x, y \in \mathbb{R}$ and $x > 0$. Then:*

$$|e^{-\alpha z} - e^{-\beta z}| \leq \left| \frac{z}{x} \right| (e^{-\alpha x} - e^{-\beta x}).$$

Proof. We can write

$$e^{-\alpha z} - e^{-\beta z} = z \int_{\alpha}^{\beta} e^{-tz} dt,$$

hence by taking absolute values:

$$|e^{-\alpha z} - e^{-\beta z}| \leq |z| \int_{\alpha}^{\beta} e^{-tx} dt = \frac{|z|}{x} (e^{-\alpha x} - e^{-\beta x}).$$

$\square$

Let $\{\lambda_n\}_{n \in \mathbb{N}}$ be an increasing sequence of real numbers tending to $+\infty$. For the sake of simplicity, we suppose that $\lambda_n \geq 0$ for every $n \in \mathbb{N}$.

Proposition 3.15. *If the series $f(z) = \sum_n a_n e^{-\lambda_n z}$ converges for $z = z_0$, it converges uniformly in every domain of the form $\Re(z - z_0) \geq 0$, $\arg(z - z_0) \leq \alpha$ with $\alpha < \pi/2$.*

Proof. After making a translation on z , we can suppose that $z_0 = 0$. The hypothesis then means that the series $\sum_n a_n$ is convergent. We must prove that there is uniform convergence in every domain of the form $\Re(z) \geq 0$, $\frac{|z|}{\Re(z)} \leq k$. Let $\varepsilon > 0$. Since the series $\sum_n a_n$ converges, there is a N such that if $m, m' \geq N$, we have $|A_{m,m'}| = \left| \sum_{n=m}^{m'} a_n \right| \leq \varepsilon$. Applying Abel's lemma on summation by parts to the sequence $e^{-\lambda_n z}$, we obtain

$$S_{m,m'} = \sum_{n=m}^{m'} a_n e^{-\lambda_n z} = \sum_{n=m}^{m'-1} A_{m,n} (e^{-\lambda_n z} - e^{-\lambda_{n+1} z}) + A_{m,m'} e^{-\lambda_{m'} z}.$$

By putting $z = x + iy$ and applying Lemma 3.14, we find

$$|S_{m,m'}| \leq \varepsilon \left(e^{-\lambda_{m'}x} + \frac{|z|}{x} \sum_{n=m}^{m'-1} (e^{-\lambda_n x} - e^{-\lambda_{n+1}x}) \right).$$

For m' sufficiently large we have $e^{-\lambda_{m'}x} \leq 1$, since $\lambda_{m'} \rightarrow +\infty$ and $e^{-\lambda_{m'}x} \rightarrow 0$ as $m' \rightarrow +\infty$. Therefore we get

$$|S_{m,m'}| \leq \varepsilon(1 + k(e^{-\lambda_m x} - e^{-\lambda_{m'}x})),$$

hence $|S_{m,m'}| \leq \varepsilon(1 + k)$, and the uniform convergence is clear. $\square$

Corollary 3.16. *If f converges for $z = z_0$, it converges for $\Re(z) > \Re(z_0)$ and the function thus defined is holomorphic.*

Proof. The proof of this fact follows from Proposition 3.15 and Lemma 3.13. $\square$

Corollary 3.17. *$f(z)$ converges to $f(z_0)$ when $z \rightarrow z_0$ in the domain*

$$\Re(z - z_0) \geq 0, \quad |\arg(z - z_0)| \leq \alpha \quad \text{with } \alpha < \pi/2.$$

Proof. This corollary follows from the uniform convergence and the fact that $e^{-\lambda_n z}$ tends to $e^{-\lambda_n z_0}$. $\square$

Corollary 3.18. *The function $f(z)$ can be identically zero if and only if all its coefficients a_n are zero.*

Proof. Let us show that a_0 is zero. We multiply f by $e^{\lambda_0 z}$ and make z tend to $+\infty$ (with z real for instance). The uniform convergence shows that $e^{\lambda_0 z}$ tends then to a_0 , hence $a_0 = 0$. We proceed similarly for a_1 and for all other coefficients. $\square$

In the case $\lambda_n = \log n$, the corresponding series is written

$$f(s) = \sum_{n=1}^{+\infty} \frac{a_n}{n^s}.$$

Proposition 3.19. *If the a_n are bounded, there is absolute convergence for $\Re(s) > 1$.*

Proof. This follows from the well known convergence of $\sum_{n=1}^{+\infty} \frac{1}{n^\alpha}$ for $\alpha > 1$. $\square$

Proposition 3.20. *If the partial sums $A_{m,p} = \sum_{n=m}^p a_n$ are bounded, there is convergence (not necessarily absolute) for $\Re(s) > 0$.*

Proof. Let us assume that $|A_{m,p}| \leq K$. By applying Abel's lemma on summation by parts, we find

$$\begin{aligned} |S_{m,m'}| &= \left| \sum_{n=m}^{m'} \frac{a_n}{n^s} \right| = \left| \sum_{n=m}^{m'-1} A_{m,n} \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right) + \frac{A_{m,m'}}{m'^s} \right| \\ &\leq K \left(\sum_{n=m}^{m'-1} \left| \frac{1}{n^s} - \frac{1}{(n+1)^s} \right| + \left| \frac{1}{m'^s} \right| \right). \end{aligned}$$

By Proposition 3.15 we can suppose that s is real. This allows us to write the previous inequality in the simpler form

$$|S_{m,m'}| \leq K/m^s,$$

and the convergence is clear. □

3.3 The L -functions

Definition 3.21. Let m be an integer ≥ 1 and let χ be a character modulo m . The corresponding L -function is defined by the Dirichlet series

$$L(s, \chi) = \sum_{n=1}^{+\infty} \frac{\chi(n)}{n^s}.$$

Let us note that, in this sum, it is only the integers n which are prime to m which give a non-zero contribution. The case of the unit character gives nothing essentially new:

Proposition 3.22. *For $\chi = 1$, we have*

$$L(s, 1) = F(s)\zeta(s) \text{ with } F(s) = \prod_{p|m} \left(1 - \frac{1}{p^s} \right).$$

In particular, $L(s, 1)$ extends analytically for $\Re(s) > 0$ and has a simple pole at $s = 1$.

Proof.

$$\begin{aligned} L(s, 1) &= \sum_{\substack{n=1 \\ (n,m)=1}}^{+\infty} \frac{1}{n^s} = \prod_{p \nmid m} \left(1 - \frac{1}{p^s} \right)^{-1} \\ &= \prod_{p|m} \left(1 - \frac{1}{p^s} \right) \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p^s} \right)^{-1} = F(s)\zeta(s). \end{aligned}$$

□

Proposition 3.23. *For $\chi \neq 1$ the series $L(s, \chi)$ converges (respectively converges absolutely) in the half plane $\Re(s) > 0$ (respectively $\Re(s) > 1$); we have*

$$L(s, \chi) = \prod_{p \in \mathbb{P}} \left(1 - \frac{\chi(p)}{p^s} \right)^{-1} \quad \text{for } \Re(s) > 1.$$

Proof. The assertion relative to $\Re(s) > 1$ follows from what has been said in Theorem 0.27. It remains to show the convergence of the series for $\Re(s) > 0$. Using Proposition 3.20, it suffices to see that the sums

$$A_{u,v} = \sum_{n=u}^v \chi(n)$$

are bounded. Now, by Proposition 3.9, we have

$$\sum_{n=u}^{u+m-1} \chi(n) = 0.$$

Hence it suffices to majorize the sums $A_{u,v}$ for $v - u < m$, and this is easily inferable: we have

$$|A_{u,v}| \leq \varphi(m).$$

The proposition follows.

□

Remark 3.24. In particular $L(1, \chi)$ is finite when $\chi \neq 1$. The essential point of Dirichlet's proof consists in showing that $L(1, \chi)$ is different from zero.

Let m be a fixed integer greater than or equal to 1. If p does not divide m , we denote by $\bar{p}$ its congruence class modulo m in $G(m) = (\mathbb{Z}/m\mathbb{Z})^*$ and by $f(p)$ the order of $\bar{p}$ in the group $G(m)$. By definition, $f(p)$ is the smallest integer $f > 1$ such that $p^f \equiv 1 \pmod{m}$. We put $g(p) = \varphi(m)/f(p)$; this is the order of the quotient of $G(m)$ by the subgroup $(\bar{p})$ generated by $\bar{p}$.

Lemma 3.25. *If $p \in \mathbb{P}$ does not divide m , we find the identity*

$$\prod_{\chi} (1 - \chi(p)T) = (1 - T^{f(p)})^{g(p)},$$

where the product extends over all characters χ of $G(m)$.

Proof. Let W be the set of $f(p)$ -th roots of unity. We have the following identity

$$\prod_{w \in W} (1 - wT) = 1 - T^{f(p)}.$$

Indeed

$$\begin{aligned} T^{f(p)} - 1 &= \prod_{w \in W} (T - w) = \prod_{w \in W} w \left(\frac{T}{w} - 1 \right) = \left(\prod_{w \in W} w \right) \prod_{w \in W} \left(\frac{T}{w} - 1 \right) \\ &\stackrel{(1)}{=} (-1)^{|W|+1} \prod_{w \in W} (w^{-1}T - 1) = (-1)^{|W|+1} \prod_{w \in W} (wT - 1), \end{aligned} \tag{3.1}$$

where in (1) we used the following fact

$$\prod_{w \in W} w = \begin{cases} 1 & \text{if } |W| \equiv 1 \pmod{2}, \\ -1 & \text{if } |W| \equiv 0 \pmod{2}. \end{cases}$$

Indeed, in the product we couple each root w with its inverse w^{-1} , except when w is equal to 1 or w is equal to -1 , but this last case happens only if $|W| \equiv 0 \pmod{2}$. Subsequently, we multiply by -1 the left- and right-hand side of (3.1) and we get

$$\begin{aligned} 1 - T^{f(p)} &= (-1)(T^{f(p)} - 1) = (-1)^{|W|+2} \prod_{w \in W} (wT - 1) \\ &= (-1)^{|W|+2} \prod_{w \in W} (-1)(1 - wT) = (-1)^{2|W|+2} \prod_{w \in W} (1 - wT) \\ &= (-1)^{2(|W|+1)} \prod_{w \in W} (1 - wT) = \prod_{w \in W} (1 - wT). \end{aligned}$$

Therefore, the claim follows from this and the fact that for all $w \in W$ there exist $g(p)$ characters χ of $G(m)$ such that $\chi(\bar{p}) = w$. Indeed the following fact holds:

Lemma 3.26. *For $(p, m) = 1$, let $f(p) = f$ be the smallest positive exponent such that $p^f \equiv 1 \pmod{m}$. Then $\chi(\bar{p})$ is a f -th root of unity and all f -th roots of unity appear equally often as $\chi(\bar{p})$ if χ runs over all characters modulo m .*

Proof. Let w be a f -th root of unity. There exists a character χ_0 such that $\chi_0(\bar{p}) = w$: indeed we define χ_0 on the subgroup generated by p and then we extend it thanks to Proposition 3.4. How are characters made such that $\chi(\bar{p}) = w$? Multiplication by χ_0^{-1} gives a bijection between the characters such that $\chi(\bar{p}) = w$ and the characters such that $\chi(\bar{p}) = 1$, that is the characters trivial on $\langle \bar{p} \rangle$. The exact sequence of Remark 3.5 shows that there are $|G(m)/\langle \bar{p} \rangle| = \varphi(m)/f$ such characters. $\square$

On the group $(\bar{p})$ there exist $f(p)$ characters. There exists one character χ of $(\bar{p})$ such that $\chi(\bar{p}) = w$ for any such root w . Any such character can be extended to $[G(m) : f(p)] = g(p)$ characters of $G(m)$ by Proposition 3.4. From this, it follows that

$$\prod_{\chi} (1 - \chi(p)T) = \left(\prod_{w \in W} (1 - wT) \right)^{g(p)} = (1 - T^{f(p)})^{g(p)}.$$

□

Definition 3.27. We define a new function $\zeta_m(s)$ by means of the formula

$$\zeta_m(s) = \prod_{\chi} L(s, \chi),$$

the product being extended over all characters χ of $G(m)$.

Proposition 3.28. *We have that*

$$\zeta_m(s) = \prod_{p \nmid m} \left(1 - \frac{1}{p^{f(p)s}} \right)^{-g(p)}.$$

This is a Dirichlet series, with positive integral coefficients, converging in the half plane $\Re(s) > 1$.

Proof. Replacing each L -function by its product expansion, and applying Lemma 3.25 (with $T = p^{-s}$), we obtain the product expansion of $\zeta_m(s)$. This expansion shows clearly that it is a series with positive coefficients; its convergence for $\Re(s) > 1$ is clear.

□

Theorem 3.29. *The following facts hold:*

- a) ζ_m has a simple pole at $s = 1$.
- b) $L(1, \chi) \neq 0$ for all $\chi \neq 1$.

Proof. If $L(1, \chi) \neq 0$ for all $\chi \neq 1$, the fact that $L(s, 1)$ has a simple pole at $s = 1$ show that the same is true for ζ_m . Thus (b) $\Rightarrow$ (a). Suppose now that $L(1, \chi) = 0$ for some $\chi \neq 1$. Then the function ζ_m would be holomorphic at $s = 1$, thus also for all s such that $\Re(s) > 0$ (Proposition 3.22 and Proposition 3.23). Since it is a Dirichlet series with positive coefficients, this series would converge for all s in the same domain, but this is absurd. Indeed, the p -th factor of ζ_m is equal to

$$\frac{1}{(1 - p^{-f(p)s})^{g(p)}} = (1 + p^{-f(p)s} + p^{-2f(p)s} + \dots)^{g(p)},$$

and dominates the series

$$1 + p^{-\varphi(m)s} + p^{-2\varphi(m)s} + \dots$$

It follows that ζ_m has all its coefficients greater than those of the series

$$\sum_{(n,m)=1} n^{-\varphi(m)s}$$

which diverges for $s = 1/\varphi(m)$. This concludes the proof. $\square$

Remark 3.30. The function ζ_m is equal (up to a finite number of factors) to the zeta function associated with the field of m -th roots of unity. The fact that ζ_m has a simple pole at $s = 1$ can also be deduced from general results on zeta functions of algebraic number fields.

We have seen that (Corollary 1.4), when s tends to 1^+ (s being real > 1), we get

$$\sum_{p \in \mathbb{P}} \frac{1}{p^s} \sim \log \left(\frac{1}{s-1} \right).$$

Recall from Definition 2.3 the notion of Dirichlet density. The theorem on arithmetic progressions can be refined in the following way:

Theorem 3.31 (Dirichlet's theorem). *Let $m \geq 1$ and let a be such that $(a, m) = 1$. Let $\mathbb{P}_a$ be the set of prime numbers such that $p \equiv a \pmod{m}$. The set $\mathbb{P}_a$ has logarithmic density $1/\varphi(m)$. (In other words the prime numbers are “equally distributed” between the different classes modulo m which are prime to m .)*

Corollary 3.32. *The set $\mathbb{P}_a$ is infinite.*

Proof. A finite set has logarithmic density zero. $\square$

Let χ be a character of $G(m)$. Set

$$f_\chi(s) = \sum_{p \nmid m} \chi(p)/p^s,$$

this series is convergent for $s > 1$.

Lemma 3.33. *If $\chi = 1$, then $f_\chi \sim \log 1/(s-1)$ for $s \rightarrow 1^+$.*

Proof. Indeed, f_1 differs from the series $\sum 1/p^s$ by a finite number of terms only. $\square$

Lemma 3.34. *If $\chi \neq 1$, f_χ remains bounded when $s \rightarrow 1^+$.*

Proof. We will use the logarithm of the function $L(s, \chi)$. It is necessary to make a little more precise what we mean by this:

$L(s, \chi)$ is defined by the product $\prod_{p \in \mathbb{P}} 1/(1 - \chi(p)p^{-s})$. For $\Re(s) > 1$ each factor is of the form $1/(1 - \alpha)$ with $|\alpha| < 1$. We define $\log 1/(1 - \alpha)$ as $\sum_{n \in \mathbb{N}} \alpha^n/n$ and we define $\log L(s, \chi)$ by the series

$$\log L(s, \chi) = \sum_{p \in \mathbb{P}} \log \left(\frac{1}{1 - \chi(p)p^{-s}} \right) = \sum_{\substack{n \in \mathbb{N} \\ p \in \mathbb{P}}} \frac{\chi(p)^n}{np^{ns}} \quad (\Re(s) > 1).$$

We now split $\log L(s, \chi)$ into two parts:

$$\log L(s, \chi) = f_\chi(s) + F_\chi(s)$$

with

$$F_\chi(s) = \sum_{p, n \geq 2} \frac{\chi(p)^n}{np^{ns}}.$$

Theorem 3.29 and Corollary 1.4 show that $\log L(s, \chi)$ and $F_\chi(s)$ remain bounded when $s \rightarrow 1^+$. Hence the same holds for $f_\chi(s)$, which proves the lemma. $\square$

Proof. (Dirichlet's theorem) We have to study the behavior of the function

$$g_a(s) = \sum_{p \in \mathbb{P}_a} \frac{1}{p^s} \quad \text{for } s \rightarrow 1^+.$$

Lemma 3.35. *We have*

$$g_a(s) = \frac{1}{\varphi(m)} \sum_{\chi} \chi(a)^{-1} f_\chi(s),$$

the sum being extended over all characters χ of $G(m)$.

Proof. The function $\sum_{\chi} \chi(a)^{-1} f_\chi(s)$ can be written, by replacing f_χ by its definition:

$$\sum_{p \nmid m} \left(\sum_{\chi} \chi(a^{-1}) \chi(p) \right) / p^s.$$

But $\chi(a^{-1}) \chi(p) = \chi(a^{-1}p)$. By Corollary 3.10, we obtain

$$\sum_{\chi} \chi(a^{-1}p) = \begin{cases} \varphi(m) & \text{if } a^{-1}p \equiv 1 \pmod{m}, \\ 0 & \text{otherwise.} \end{cases}$$

Hence we find the function $\varphi(m)g_a(s)$. $\square$

Dirichlet's theorem is now clear. Indeed, Lemma 3.33 shows that $f_\chi(s) \sim \log 1/(s-1)$ for $\chi = 1$, and Lemma 3.34 shows that all other f_χ remain bounded. Using Lemma 3.35, we then see that $g_a(s) \sim (1/\varphi(m))(\log 1/(s-1))$, and this means that the density of $\mathbb{P}_a$ is $1/\varphi(m)$, so the claim follows. $\square$

Remark 3.36. The proof shown of Dirichlet's theorem also provides information about the logarithmic density, which is equal to $1/\varphi(m)$, of the set $\mathbb{P}_a$ of primes numbers such that $p \equiv a \pmod{m}$. A priori, nothing is known about the natural density of $\mathbb{P}_a$. Actually, Dirichlet's theorem is also true for natural density. To prove the version with natural density, it is possible retrace a path similar to that taken in Chapter 1 for the prime number theorem. Also in this case the hardest part is to show that the functions $L(s, \chi)$ do not vanish for $\Re(s) = 1$. For a proof of this fact see [1].

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